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CONNECTING STRUCTURE, CHILD CARRYING DEVICE, FRAME, AND CHILD CARRIER

Abstract

A connecting structure is provided, including: a first joint component and a second joint component pivotable relative to each other around an axial direction; and a connecting member extending through the second joint component in the axial direction and inserted into and screwedly connected to the first joint component to restrict a separation of the first joint component from the second joint component in a direction parallel to the axial direction. The connecting member is spaced apart from the second joint component.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priorities of China Patent Application No. 202410245771.0 and filed on Mar. 4, 2024, China Patent Application No. 202410888301.6 and filed on Jul. 3, 2024, and China Patent Application No. 202411253488.9 and filed on Sep. 6, 2024, the contents of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a connecting structure, a child carrying device, a frame, and a child carrier.

BACKGROUND

[0003] Due to their convenience for storage and transport, the foldable strollers in the market are widely favored by consumers. However, there is still room for improvement in current strollers to meet the needs of modern families and enhance the user's experience.

SUMMARY

[0004] In a first aspect, the present disclosure provides a connecting structure, including: a first joint component and a second joint component pivotable relative to each other around an axial direction; and a connecting member extending through the second joint component in the axial direction and inserted into and screwedly connected to the first joint component to restrict the separation of the first joint component from the second joint component in a direction parallel to the axial direction. The connecting member is spaced apart from the second joint component.

[0005] In an embodiment, the first joint component is positioned between the connecting member and the second joint component in a radial direction perpendicular to the axial direction to space apart the connecting member from the second joint component in the radial direction.

[0006] In an embodiment, the first joint component includes a shaft portion extending through the second joint component in the direction parallel to the axial direction. The shaft portion is positioned between the connecting member and the second joint component in the radial direction.

[0007] In an embodiment, the connecting member is provided with a head portion and a rod portion extending from the head portion in the axial direction. The rod portion is inserted into the shaft portion along the axial direction and threadedly connected to the shaft portion. The head portion is positioned at an outer side of the shaft portion in the axial direction. The second joint component is positioned between the head portion and the first joint component in the direction parallel to the axial direction.

[0008] In an embodiment, the connecting member further includes a spacer member. The connecting member is provided with a head portion and a rod portion extending from the head portion in the axial direction. The rod portion is inserted into the shaft portion along the axial direction and threadedly connected to the shaft portion. The head portion is positioned at an outer side of the shaft portion in the axial direction. The spacer member is positioned between the head portion and the second joint component in the direction parallel to the axial direction to space apart the connecting member from the second joint component in the direction parallel to the axial direction. The spacer member is unable to rotate relative to the connecting member around the axial direction.

[0009] In an embodiment, the spacer member is sleeved outside of the shaft portion along the direction parallel to the axial direction, and the spacer member is unable to rotate relative to the shaft portion around the axial direction.

[0010] In an embodiment, an inner peripheral surface of the spacer member is positioned adjacent

to an outer surface of the shaft portion in the radial direction. A shape of a cross section of the inner peripheral surface of the spacer member taken along a direction perpendicular to the axial direction and a shape of a cross section of the outer surface of the shaft portion taken along the direction perpendicular to the axial direction are matched with each other and are non-circular.

[0011] In an embodiment, the head portion, the spacer member, the second joint component, and the first joint component abut against each other sequentially in the direction parallel to the axial direction.

[0012] In an embodiment, the second joint component is provided with a second recess that is recessed away from the head portion in the direction parallel to the axial direction. The spacer member is received in the second recess along the direction parallel to the axial direction. The second recess includes a second bottom surface. The spacer member is positioned between the head portion and the second bottom surface in the direction parallel to the axial direction.

[0013] In an embodiment, the second recess includes an inner surface positioned adjacent to and spaced apart from an outer peripheral surface of the spacer member in the radial direction.

[0014] In an embodiment, the spacer member is positioned between the head portion and the shaft portion in the direction parallel to the axial direction and abuts against each of the head portion and the shaft portion.

[0015] In an embodiment, an end portion of the shaft portion proximate to the head portion is recessed away from the head portion in the direction parallel to the axial direction to form a first recess. The spacer member is received in the first recess along the direction parallel to the axial direction. The first recess includes a first bottom surface. The spacer member is positioned between the head portion and the first bottom surface in the direction parallel to the axial direction.

[0016] In an embodiment, the first recess includes an outer surface. The inner peripheral surface of the spacer member is adjacent to the outer surface in the radial direction. The shapes of the cross sections of the inner peripheral surface of the spacer member and the outer surface taken along a direction perpendicular to the axial direction are matched with each other and are non-circular.

[0017] In an embodiment, the connecting structure further includes a spacer member. The spacer member is positioned between the connecting member and the second joint component in the direction parallel to the axial direction to space apart the connecting member from the second joint component in the direction parallel to the axial direction. The spacer member is unable to rotate relative to the connecting member around the axial direction.

[0018] In an embodiment, the spacer member is sleeved outside of the first joint component along the direction parallel to the axial direction, and the spacer member is unable to rotate relative to the first joint component around the axial direction.

[0019] In an embodiment, the inner peripheral surface of the spacer member is positioned adjacent to the outer surface of the first joint component in the radial direction. The shapes of the cross sections of the inner peripheral surface of the spacer member and the outer surface of the first joint component taken along a direction perpendicular to the axial direction are matched with each other and are non-circular.

[0020] In an embodiment, the spacer member abuts against each of the connecting member and the second joint component in the direction parallel to the axial direction. The second joint component is positioned between the connecting member and the first joint component in the direction parallel to the axial direction. The second joint component abuts against the first joint component.

[0021] In an embodiment, the spacer member is positioned between the connecting member and the first joint component in the direction parallel to the axial direction, and abuts against each of the connecting member and the first joint component.

[0022] In an embodiment, the connecting member further includes a locking member, a reset member, and an unlocking member. The locking member is disposed between the first joint component and the second joint component in the direction parallel to the axial direction, and is capable of moving between a locking position and an unlocking position along the direction

parallel to the axial direction. In the locking position, the locking member prevents the first joint component and the second joint component from rotating relative to each other. In the unlocking position, the locking member allows the first joint component and the second joint component to rotate relative to each other. The reset member is disposed between the locking member and the first joint component and configured to bias the locking member toward the locking position. The unlocking member is movably connected to the second joint component, and is configured to be operable externally from the connecting structure to drive the locking member to move from the locking position to the unlocking position. The locking member is positioned between the reset member and the unlocking member in the direction parallel to the axial direction.

[0023] In an embodiment, the connecting member is exposed outside of the first joint component and the second joint component in the axial direction. The connecting member is positioned between the unlocking member and the first joint component and between the unlocking member and the second joint component in the direction parallel to the axial direction.

[0024] In an embodiment, the connecting member further includes a reinforcing member disposed inside the first joint component.

[0025] The present disclosure further provides a child carrier, including: a traveling mechanism; a lower handlebar extending upward from the traveling mechanism; an upper handlebar extending upward from the lower handlebar; and the connecting structure according to the first aspect of the present disclosure. The first joint component is fixedly connected to the lower handlebar. The second joint component is fixedly connected to the upper handlebar. In a second aspect, the present disclosure provides a child carrying device. The child carrying device includes a folding joint and an operation assembly. The folding joint includes a mounting base, a first pivoting base, and a second pivoting base pivotably connected to each other by a first pivot shaft. The mounting base is configured to be mounted to a connecting joint of a child carrier. The first pivoting base is disposed between the mounting base and the second pivoting base. The operation assembly includes an operation member and a folding unlocking member. The folding unlocking member is pivotably connected to the folding joint via the first pivot shaft, and is operably connected to the folding joint. The operation member is a rigid structure and is pivotably connected to the folding unlocking member via a second pivot shaft. The second pivot shaft is offset from the first pivot shaft. The operation member is adapted to drive the folding unlocking member to rotate to drive the folding joint to switch from a locked state to an unlocked state, thereby allowing the first pivoting base to rotate relative to the mounting base.

[0026] In an embodiment, the operation member is provided with an operation portion adapted to be operated. The folding unlocking member has a first position. When the folding unlocking member is in the first position, the folding joint is in the locked state, and the operation portion of the operation member is positioned above a plane where the first pivot shaft is located and/or above a plane where the second pivot shaft is located.

[0027] In an embodiment, the folding unlocking member has a first position. When the folding unlocking member is in the first position, the folding joint is in the locked state, and a first angle between a plane defined collectively by the first pivot shaft and the second pivot shaft and the vertical direction is an acute angle.

[0028] In an embodiment, the folding unlocking member has a second position. When the folding unlocking member is in the second position, the folding joint is unlocked, and a second angle between a plane defined collectively by the first pivot shaft and the second pivot shaft and the vertical direction is an acute angle. The second angle is smaller than the first angle.

[0029] In an embodiment, when the folding unlocking member is rotated to the second position by the operation portion of the operation member subjected to a pulling force, a plane defined collectively by the operation portion and the second pivot shaft is parallel to a direction of the pulling force and the direction of the pulling force is vertically upward.

[0030] In an embodiment, the folding unlocking member has a second position. As the folding

unlocking member rotates from the first position to the second position around the first pivot shaft, the folding unlocking member rotates relative to the second pivoting base to unlock the folding joint.

[0031] In an embodiment, under action the pulling force, the operation member drives the folding unlocking member to rotate from the second position to a third position around the first pivot shaft via the second pivot shaft. When the folding unlocking member is in the third position, the plane defined collectively by the first pivot shaft and the second pivot shaft is parallel to the direction of the pulling force and the pulling force is directed vertically upward.

[0032] In an embodiment, during the rotation of the folding unlocking member from the second position to the third position, the plane defined collectively by the operation portion of the operation member and the second pivot shaft is parallel to the direction of the pulling force.

[0033] In an embodiment, the folding unlocking member is provided with a limiting recess. The operation member is inserted into the limiting recess and pivotably connected to a wall of the limiting recess via the second pivot shaft.

[0034] In an embodiment, the wall of the limiting recess includes a first side wall and a second side wall located at two sides of the second pivot shaft, respectively. The operation member is adapted to selectively abut against the first side wall and the second side wall to limit the position of the operation member relative to the folding unlocking member.

[0035] In an embodiment, the first pivot shaft and the second pivot shaft collectively define a first plane. The second side wall is inclined to the first plane. A distance between the second side wall and the first plane decreases from a position close to an opening end of the limiting recess to a position away from the opening end of the limiting recess.

[0036] In the second aspect, the present disclosure further provides a child carrier, including: a supporting device provided with a connecting joint; and the child carrying device in the second aspect. The folding joint of the child carrying device is operably connected to the connecting joint. The folding joint is capable of unlocking the connecting joint in its unlocking process to allow the supporting device to be folded.

[0037] In a third aspect, the present disclosure provides a frame switchable between an unfolded state and a folded state, including: a first frame body, a second frame body, a rotation member, and a buffer member. The second frame body is pivotally connected to the first frame body. As the second frame body rotates relative to the first frame body in a first rotation direction, the frame switches from the unfolded state to the folded state. The rotation member is pivotally connected to the first frame body. The buffer member is connected to the rotation member. The buffer member is configured to provide a force to the rotation member for rotation relative to the first frame body in a second rotation direction opposite to the first rotation direction. During the rotation of the second frame body relative to the first frame body in the first rotation direction, the second frame body drives the rotation member to rotate relative to the first frame body in the first rotation direction.

[0038] In an embodiment, the first frame body has a first edge and a second edge. The second frame body has a third edge and a fourth edge. A first angle is formed between the first edge and the third edge. A value of the first angle when the frame is in the unfolded state is greater than a value of the first angle when the frame is in the folded state. The rotation member is positioned in an area between the first edge and the third edge. Additionally or alternatively, during the rotation of the second frame body relative to the first frame body in the first rotation direction, the rotation member abuts against the third edge of the second frame body.

[0039] In an embodiment, there is a gap at a pivot position between the first frame body and the second frame body, and the gap is shielded by the rotation member.

[0040] In an embodiment, the buffer member is a resilient member. Alternatively, the buffer member is a torsion spring, the rotation member has an accommodation groove, one of two torsion arms of the buffer member is connected to the first frame body, and the other one of the two torsion arms is positioned in the accommodation groove.

[0041] In an embodiment, the first frame body includes a first main frame body and a first connecting base connected to each other and arranged at an angle. The second frame body includes a second main frame body and a second connecting base connected to each other. The first connecting base is pivotally connected to the second connecting base.

[0042] In an embodiment, when the frame is in the unfolded state, the rotation member abuts against an edge of the first connecting base facing the rotation member or an edge of the second connecting base facing the rotation member.

[0043] In an embodiment, during switching of the frame from the unfolded state to the folded state, the edge of the second connecting base facing the rotation member pushes the rotation member to rotate relative to the first frame body in the first rotation direction.

[0044] In an embodiment, the first connecting base includes a first base body and a second base body spaced apart from each other. The first connecting base has a cavity and an opening in communication with each other. The cavity is defined between the first base body and the second base body. At least a part of the second connecting base is embedded into the cavity. The opening is disposed opposite to the rotation member. The third edge of the second frame body is exposed through the opening. During switching of the frame from the unfolded state to the folded state, the third edge protrudes out of the opening and pushes the rotation member to rotate relative to the first frame body in the first rotation direction.

[0045] In an embodiment, the rotation member has a width greater than a distance between the first frame body and the second frame body, and at least a part of the opening is shielded by the rotation member.

[0046] In an embodiment, the frame further includes a first pivot shaft. The first connecting base is pivotally connected to the second connecting base via the first pivot shaft. The first connecting base has a first portion from a connection position between the first connecting base and the first main frame body to the first pivot shaft, and a second portion from the first pivot shaft to an end of the first connecting base. When the frame is in the unfolded state, an area of the opening corresponding to the first portion is shielded by the rotation member.

[0047] In an embodiment, the first connecting base further includes a connecting portion connecting the first base body to the second base body, and the cavity is enclosed by the connecting portion, the first base body, and the second base body.

[0048] In an embodiment, the first frame body is a handlebar frame, and the second frame body is a rear leg frame. The frame further includes a front leg frame pivotally connected to the handlebar frame.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0049] FIG. 1 is a perspective view of a stroller provided according to a first embodiment of the present disclosure;

[0050] FIG. 2A and FIG. 2B are partially enlarged views of the boxes at the left side and the right side of FIG. 1, respectively;

[0051] FIG. 3A and FIG. 3B are exploded perspective views corresponding to FIG. 2A and FIG. 2B, respectively;

[0052] FIG. 4 is a perspective view of a connecting structure according to the first embodiment of the present disclosure, with the unlocking member removed;

[0053] FIG. 5 is a perspective view of the connecting structure according to the first embodiment of the present disclosure, with the unlocking member, the connecting member, and the spacer member removed;

[0054] FIG. 6 is a perspective view of the connecting structure according to the first embodiment

of the present disclosure, with the unlocking member and the connecting member removed;

[0055] FIG. 7 is a perspective view of the connecting structure according to the first embodiment of the present disclosure, with the unlocking member, the second joint component, and the spacer member removed;

[0056] FIG. 8 is a side cross-sectional view of the connecting structure according to the first embodiment of the present disclosure;

[0057] FIG. 9 is a partially enlarged view of the box at the spacer member of FIG. 8;

[0058] FIG. 10 is a schematic perspective view of a child carrier provided according to a second embodiment of the present disclosure, with a child carrying device in the folded state, a supporting device in the unfolded state, and a folding joint in the locked state;

[0059] FIG. 11 is a schematic cross-sectional view of the child carrier taken along the U1-U1 line of FIG. 10, with the child carrying device in the folded state, the supporting device in the unfolded state, and the folding joint in the locked state;

[0060] FIG. 12 is a schematic cross-sectional view of the child carrier taken along the U1-U1 line of FIG. 10, with the child carrying device in the folded state, the supporting device in the unfolded state, and the folding joint in the unlocked state;

[0061] FIG. 13 is a schematic cross-sectional view of the child carrier taken along the U1-U1 line of FIG. 10, with both the child carrying device and the supporting device in the folded state and the folding joint in the unlocked state;

[0062] FIG. 14 is a schematic cross-sectional view of the child carrier taken along the U1-U1 line of FIG. 10, with both the child carrying device and the supporting device in the folded state and the folding joint in the unlocked state;

[0063] FIG. 15 is a schematic cross-sectional view of the child carrier taken along the U2-U2 line of FIG. 11, with a folding locking member in the locking position;

[0064] FIG. 16 is a schematic cross-sectional view of the child carrier taken along the U3-U3 line of FIG. 12, with a folding locking member in the unlocking position;

[0065] FIG. 17 is a schematic perspective view of the child carrying device of FIG. 10, with the folding locking member in the unlocking position;

[0066] FIG. 18 is a schematic partial sectional view of the child carrying device taken along the U4-U4 line of FIG. 17, with the folding locking member in the locking position;

[0067] FIG. 19 is a schematic partial exploded view of the folding joint of the child carrying device of FIG. 18 in one direction;

[0068] FIG. 20 is a schematic partial exploded view of the folding joint of the child carrying device of FIG. 18 in another direction;

[0069] FIG. 21 is a schematic partial exploded view of the connecting joint of the child carrying device of FIG. 18 in one direction;

[0070] FIG. 22 is a schematic partial exploded view of the connecting joint of the child carrying device of FIG. 18 in another direction;

[0071] FIG. 23 is a schematic perspective view of a frame in the unfolded state according to a third embodiment of the present disclosure;

[0072] FIG. 24 is a partial exploded view of the frame in the unfolded state according to the third embodiment of the present disclosure;

[0073] FIG. 25 is a cross-sectional view of the frame in the unfolded state according to the third embodiment of the present disclosure;

[0074] FIG. 26 is a cross-sectional view of the frame at a time point during the folding process according to the third embodiment of the present disclosure;

[0075] FIG. 27 is a side view of the frame in the folded state according to the third embodiment of the present disclosure;

[0076] FIG. 28 is a cross-sectional view of the frame in the folded state according to the third embodiment of the present disclosure.

DETAILED DESCRIPTION

[0077] The present disclosure will now be described in detail with reference to the accompanying drawings and embodiments in order to make the objects, technical solutions, and advantages of the present disclosure more clear. In the following description, numerous specific details are set forth in order to provide a full understanding of the present disclosure. However, the present disclosure may be implemented otherwise than as described herein, and similar modifications may be made by those skilled in the art without departing from the scope of the present disclosure. The following specific embodiments are not intended to limit the scope of the present disclosure.

[0078] In the description of the present disclosure, it is to be understood that the orientation and position relationships indicated by the technical terms such as “central”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “upper”, “lower”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, “clockwise”, “anticlockwise”, “axial”, “radial”, “circumferential”, etc., are based on the orientation or position relationship as shown in the accompanying drawings, it is only for the convenience of describing the embodiments of the present disclosure and simplifying the description, rather than indicating or implying that the device or element referred to thereby must have a specific orientation and be constructed and operated in a specific orientation, and thus it cannot be understood as limitations on the embodiments of the present disclosure.

[0079] In addition, terms such as “first” and “second” are used herein for purposes of description and are not intended to indicate or imply relative importance or significance or to imply the number of indicated technical features. Thus, features limited by “first” and “second” are intended to indicate or imply including one or more than one of these features. Further, in the description of the present disclosure, “a plurality of” means two or more than two, unless specified otherwise.

[0080] It is to be understood that when an element is referred to as being “fixed” or “disposed” on another element, it can be directly on the other element or intervening elements may be present. When an element is referred to as being “connected” to another element, it can be directly connected to the other element or intervening elements may be present.

First Embodiment

[0081] A stroller typically includes a traveling mechanism, such as a wheel, and a handlebar positioned above the traveling mechanism. The user pushes the handlebar to move the stroller. Many strollers currently available on the market include a folding mechanism or height adjustment mechanism for the handlebar, wherein the rod portions of the handlebar are connected by two or more joints that are rotatable relative to each other.

[0082] A significant issue arises when using a bolt and a nut to connect these joints of the handlebar. In this configuration, the bolt passes through all the joints simultaneously, with both ends thereof exposed out of the joints, wherein one end of the bolt presses against one of the joints, while the other end of the bolt is threadedly connected to the nut to connect the joints together. However, when the joints rotate relative to each other, they also rotate relative to the bolt and tends to apply a rotational torque to the bolt, which may lead to gradual loosening of the bolt and the nut.

[0083] Therefore, there is a need to provide a new connecting structure that prevents the joints from applying a rotational torque to the bolt during their relative rotation. It is also desirable that the connecting structure provides a neat appearance, particularly by concealing the bolt and the nut. Additionally, it is also desirable for the connecting structure to incorporate a locking device in a compact manner to lock or unlock the relative rotation between the joints.

[0084] To address the above issues, the first embodiment of the present disclosure provides a connecting structure and a stroller including the same, as shown in FIG. 1 to FIG. 9.

[0085] The stroller **1000** according to the first embodiment of the present disclosure will be generally described with reference to FIG. 1. The stroller **1000** includes a traveling mechanism **1400**, a lower handlebar **1200**, an upper handlebar **1300**, a connecting structure **1100**, a front leg **1500**, and a rear leg **1600**. The traveling mechanism **1400** includes, for example, a wheel, which is

disposed at the bottom of the stroller **1000**. The front leg **1500** and the rear leg **1600** each extend upward from the traveling mechanism **1400**. The lower handlebar **1200** may be connected to the traveling mechanism **1400** via the rear leg **1600**, but the present disclosure is not limited thereto. The upper handlebar **1300** extends upward from the lower handlebar **1200**. The connecting structure **1100** is connected between the upper handlebar **1300** and the lower handlebar **1200** such that the upper handlebar **1300** and the lower handlebar **1200** are capable of rotating relative to each other around a transverse axial direction A. In other embodiments, the connecting structure **1100** may be provided between the lower handlebar **1200** and the front leg **1500**, between the lower handlebar **1200** and the rear leg **1600**, or between the front leg **1500** and the rear leg **1600** of the stroller **1000**.

[0086] Referring to FIG. 2A and FIG. 2B, the connecting structure **1100** according to the first embodiment of the present disclosure includes a first joint component **1110** and a second joint component **1120**. The first joint component **1110** is fixedly connected to the lower handlebar **1200**, the second joint component **1120** is fixedly connected to the upper handlebar **1300**, and the first joint component **1110** and the second joint component **1120** are capable of rotating relative to each other around the axial direction A. As such, the relative rotation between the upper handlebar **1300** and the lower handlebar **1200** are achieved via the connecting structure **1100**.

[0087] In the first embodiment of the present disclosure, two lower handlebars **1200** parallel to each other may be provided. The upper handlebar **1300** may be substantially U-shaped. The two ends of the upper handlebar **1300** are rotatably connected to the two lower handlebars **1200** by two connecting structures **1100**, respectively. The two connecting structures **1100** are disposed mirror-symmetrically opposite to one another along the axial direction A.

[0088] The connecting structure **1100** according to the first embodiment of the present disclosure will be described in detail with reference to FIG. 3A and FIG. 3B. The connecting structure **1100** includes a first joint component **1110**, a second joint component **1120**, a connecting member **1130** (e.g., a screw, with the end of the screw provided with a groove), a spacer member **1140**, a locking member **1150**, an unlocking member **1160**, a reset member **1170**, and a reinforcing member **1180**.

[0089] The connecting structure **1100** according to the first embodiment of the present disclosure will be further described with reference to FIG. 3A and FIG. 8. In general, the first joint component **1110** includes a first disk-shaped portion **1111**. The second joint component **1120** includes a second disk-shaped portion **1121**. The first disk-shaped portion **1111** and the second disk-shaped portion **1121** are connected to each other and are rotatable relative to each other around the axial direction A. The connecting member **1130** extends through the second joint component **1120** along the axial direction A to be screwedly connected (specifically threaded) to the first joint component **1110**, such that the connecting member **1130** is fixed relative to the first joint component **1110**. The connecting member **1130** does not extend through the first joint component **1110** along the axial direction A, with only one end of the connecting member **1130** in the axial direction A exposed outside of the first joint component **1110**. At least a part of the second joint component **1120** is positioned between the connecting member **1130** and the first joint component **1110** in a direction parallel to the axial direction A, such that the second joint component **1120** is restrained between the connecting member **1130** and the first joint component **1110** in the direction parallel to the axial direction A, thereby preventing the separation of the first joint component **1110** from the second joint component **1120** in the axial direction A. The end of the connecting member **1130** exposed outside of the first joint component **1110** is also exposed outside of the second joint component **1120** (specifically see FIG. 4).

[0090] During the rotation of the second disk-shaped portion **1121** relative to the first disk-shaped portion **1111**, the second disk-shaped portion **1121** may apply a rotational torque to the connecting member **1130** threadedly connected to the first disk-shaped portion **1111**, causing the connecting member **1130** to rotate relative to the first disk-shaped portion **1111**, which may in turn cause the separation of the connecting member **1130** from the first disk-shaped portion **1111**. To avoid this,

the second disk-shaped portion **1121** of the present disclosure is spaced apart from the connecting member **1130**, so that the application of the rotational torque from the second disk-shaped portion **1121** to the connecting member **1130** is avoided while retaining the basic function of the connecting member **1130** of restricting the separation of the first joint component **1110** from the second joint component **1120** in the axial A direction.

[0091] The connecting structure **1100** according to the first embodiment of the present disclosure will be further described with reference to FIG. 3A, FIG. 5, and FIG. 8.

[0092] The first disk-shaped portion **1111** includes a shaft portion **1112** extending toward the second joint component **1120** in the direction parallel to the axial direction A. The shaft portion **1112** extends from the center of the first disk-shaped portion **1111** and through the center of the second disk-shaped portion **1121** in the direction parallel to the axial direction A to be exposed outside of the second disk-shaped portion **1121**. That is, the second joint component **1120** is sleeved outside of the shaft portion **1112** such that the second disk-shaped portion **1121** can rotate relative to the first disk-shaped portion **1111** around the shaft portion **1112**.

[0093] The connecting structure **1100** according to the first embodiment of the present disclosure will be further described with reference to FIG. 7 and FIG. 8.

[0094] The connecting member **1130** is inserted into the shaft portion **1112** from the side of the second disk-shaped portion **1121** facing away from the first joint component **1110** (i.e., the right side in FIG. 3A) along the axial direction A, and threadedly connected to the shaft portion **1112**, but does not extend through the shaft portion **1112** along the axial direction A. It should be appreciated that in this case, the connecting member **1130** extends through the second disk-shaped portion **1121**, and the shaft portion **1112** is positioned between the connecting member **1130** and the second joint component **1120** in a radial direction perpendicular to the axial direction A to space apart the connecting member **1130** from the second joint component **1120** in the radial direction. As such, the application of the rotational torque from the second joint component **1120** to the connecting member **1130** in the radial direction during the rotation of the second joint component **1120** relative to the first joint component **1110** can be avoided.

[0095] Specifically, the shaft portion **1112** is provided with a shaft hole **1112b** therein for connecting with the connecting member **1130**. For example, the shaft hole **1112b** may be provided with a female thread, and the connecting member **1130** may be provided with a male thread, such that the connecting member **1130** can be threadedly fixed into the shaft hole **1112b**.

[0096] Specifically, the connecting member **1130** includes a head portion **1131** and a rod portion **1132**. The rod portion **1132** is inserted into the shaft portion **1112** along the axial direction A and threadedly connected to the shaft portion **1112**, such that the connecting member **1130** is connected to the first joint component **1110**. The head portion **1131** is positioned at an outer side of the shaft portion **1112** in the axial direction A, and at least a part of the second joint component **1120** is positioned between the head portion **1131** and the first joint component **1110** in the direction parallel to the axial direction A, such that the second joint component **1120** is restricted between the connecting member **1130** and the first joint component **1110** in the direction parallel to the axial direction A, thereby preventing the separation of the first joint component **1110** from the second joint component **1120** along the axial direction A. Additionally, the head portion **1131** of the connecting member **1130** is exposed outside of the first joint component **1110** and the second joint component **1120** in the axial direction A (see FIG. 4).

[0097] When at least a part of the second joint component **1120** is positioned between the head portion **1131** and the first joint component **1110** in the direction parallel to the axial direction A, the at least a part of the second joint component **1120** is positioned opposite to the head portion **1131** in the direction parallel to the axial direction A. To avoid the abutment of the connecting member **1130** against the second joint component **1120** in the direction parallel to the axial direction A, the head portion **1131** is spaced apart from the second joint component **1120** in the direction parallel to the axial direction A. Specifically, in the present disclosure, the spacer member **1140** is disposed

between the at least a part of the second joint component **1120** and the head portion **1131** in the direction parallel to the axial direction A (see FIG. 8) such that the second joint component **1120** is spaced apart from the connecting member **1130** (specifically, the head portion **1131**), and the spacer member **1140** is unable to rotate relative to the connecting member **1130** around the axial direction A.

[0098] The connecting structure **1100** according to the first embodiment of the present disclosure will be further described with reference to FIG. 8. Since the at least a part of the second joint component **1120** is positioned between the spacer member **1140** and the first joint component **1110** in the direction parallel to the axial direction A, it should be appreciated that the head portion **1131**, the spacer member **1140**, the at least a part of the second joint component **1120**, and at least a part of the first joint component **1110** are arranged sequentially in the direction parallel to the axial direction A to prevent the separation of the second joint component **1120** from the first joint component **1110** in the direction parallel to the axial direction A.

[0099] In other embodiments, the connecting member **1130** may be in other forms, such as a bolt (which does not need to be connected to a nut, but is only threadedly connected to the first joint component **1110**, with no groove on the end thereof), as long as the separation of the second joint component **1120** from the first joint component **1110** in the axial direction A can be prevented.

[0100] In the first embodiment of the present disclosure, the spacer member **1140** is unable to rotate relative to the connecting member **1130** around the axial direction A, specifically relative to the first joint component **1110**, and the first joint component **1110** is fixed relative to the connecting member **1130**.

[0101] It should be appreciated from the above that the head portion **1131**, the spacer member **1140**, the second joint component **1120**, and the first joint component **1110** are arranged sequentially in the direction parallel to the axial direction A. That is, the second joint component **1120** is arranged adjacent to the spacer member **1140**, so that during the rotation of the second joint component **1120** relative to the first joint component **1110**, the second joint component **1120** may apply a rotational torque to the spacer member **1140**. To prevent the second joint component **1120** from applying a rotational torque to the spacer member **1140** and then applying a rotational torque to the head portion **1131** via the spacer member **1140**, in the present disclosure, the spacer member **1140** is sleeved onto a radially outer side of the first joint component **1110**, as shown in FIG. 6 and FIG. 8. Specifically, the spacer member **1140** is sleeved onto a radially outer side of the shaft portion **1112** of the first joint component **1110**, and the spacer member **1140** is unable to rotate relative to the shaft portion **1112** around the axial direction A. Referring to FIG. 9, the spacer member **1140** has an outer peripheral surface **1142** and an inner peripheral surface **1141**. The inner peripheral surface **1141** of the spacer member **1140** is positioned at an outer side of the outer surface **1114b** of the shaft portion **1112** of the first joint component **1110**. The shapes of the cross sections of the inner peripheral surface **1141** and the outer surface **1114b** taken along a direction perpendicular to the axial direction A are matched with each other (e.g., substantially the same) and are non-circular, so as to prevent the rotation of the inner peripheral surface **1141** relative to the outer surface **1114b** around the axial direction A. Referring to FIG. 9, in this case, the inner peripheral surface **1141** and the outer surface **1114b** may not fully abut against each other in the radial direction. Referring to FIG. 8, to ensure that the spacer member **1140** and the second joint component **1120** will not move between the head portion **1131** and the first joint component **1110** in the direction parallel to the axial direction A, the head portion **1131**, the spacer member **1140**, the second joint component **1120**, and the first joint component **1110** sequentially abut against each other in the direction parallel to the axial direction A. In other embodiments, the spacer member **1140** may not abut against the head portion **1131** in the direction parallel to the axial direction A, or the second joint component **1120** may not abut against the first joint component **1110** in the direction parallel to the axial direction A, so that the spacer member **1140** and the second joint component **1120** may sway slightly between the head portion **1131** and the first joint component **1110**.

[0102] In order to enable the spacer member **1140** to be more stably positioned between the head portion **1131** and the second joint component **1120** in the direction parallel to the axial direction A, referring to FIG. 8, the spacer member **1140** abuts against each of the head portion **1131** and at least a part of the first joint component **1110** in the direction parallel to the axial direction A to further fix the position of the spacer member **1140** in the direction parallel to the axial direction A. In other embodiments, the second joint component **1120** may not abut against the spacer member **1140** or the first joint component **1110**, so that the second joint component **1120** may sway slightly between the spacer member **1140** and the first joint component **1110** in the direction parallel to the axial direction A.

[0103] In other embodiments, the shapes of the cross sections of the inner peripheral surface **1141** and the outer surface **1114b** taken along a direction perpendicular to the axial direction A may be matched with each other and be circular, and the rotation of the spacer member **1140** relative to the shaft portion **1112** around the axial direction A may be restricted by the friction force between the inner peripheral surface **1141** and the outer surface **1114b**.

[0104] The connection relationship between the spacer member **1140** and the connecting member **1130**, the first joint component **1110**, and the second joint component **1120** will be described in detail in the below.

[0105] Referring to FIG. 5, FIG. 6, FIG. 8, and FIG. 9, an end portion of the shaft portion **1112** proximate to the head portion **1131** is recessed away from the head portion **1131** in the direction parallel to the axial direction A to form a first recess **1114** around the axial direction A located at an outer periphery of the shaft portion **1112**. The spacer member **1140** is received in the first recess **1114** along the direction parallel to the axial direction A. The first recess **1114** includes a first bottom surface **1114a** and an outer surface **1114b** (the outer surface **1114b** of the first recess **1114** is a part of the outer surface of the shaft portion **1112**). The extending direction of the first bottom surface **1114a** is perpendicular to the axial direction A. The outer surface **1114b** is disposed around the axial direction A and extends along the direction parallel to the axial direction A. The spacer member **1140** is positioned between the head portion **1131** and the first bottom surface **1114a** in the direction parallel to the axial direction A, and abuts against each of the head portion **1131** and the first bottom surface **1114a**, such that the spacer member **1140** is sandwiched between the head portion **1131** and the first joint component **1110** in the direction parallel to the axial direction A. The outer diameter of the spacer member **1140** (i.e., the diameter of the outer peripheral surface **1142**) is larger than the outer diameter of the head portion **1131** of the connecting member **1130**, so that the portion of the spacer member **1140** extending beyond the outer diameter of the head portion **1131** in the radial direction can be used to abut against the second joint component **1120** in the direction parallel to the axial direction A.

[0106] Referring to FIG. 5 and FIG. 6, the cross sections of the inner peripheral surface **1141** and the outer surface **1114b** taken along a direction perpendicular to the axial direction A each include four arcs **1141a** protruded outwardly that are arranged along the circumferential direction to restrict the rotation of the spacer member **1140** relative to the first joint component **1110** around the axial direction A. In other embodiments, the cross sections of the inner peripheral surface **1141** and the outer surface **1114b** taken along a direction perpendicular to the axial direction A may have a square shape, a hexagonal shape, a keyway shape, or other shapes, as long as the rotation of the spacer member **1140** relative to the first joint component **1110** around the axial direction A can be prevented.

[0107] Referring to FIG. 5, FIG. 6, FIG. 8, and FIG. 9, the second joint component **1120** is provided with a second recess **1123** that is recessed away from the head portion **1131** in the direction parallel to the axial direction A. The second recess **1123** is arranged surrounding the shaft portion **1112** and is located at a radially inner side of the second joint component **1120**. The spacer member **1140** is received in the second recess **1123** along the direction parallel to the axial direction A. The second recess **1123** includes a second bottom surface **1123a** and an inner surface **1123b**.

The extending direction of the second bottom surface **1123a** is perpendicular to the axial direction A. The inner surface **1123b** is arranged around the axial direction A and extends along the direction parallel to the axial direction A. The spacer member **1140** is positioned between the head portion **1131** and the second bottom surface **1123a** in the direction parallel to the axial direction A and abuts against each of the head portion **1131** and the second bottom surface **1123a**, such that the spacer member **1140** is sandwiched between the head portion **1131** and the second joint component **1120** in the direction parallel to the axial direction A.

[0108] Referring to FIG. 6, FIG. 8, and FIG. 9, the spacer member **1140** is also positioned between the inner surface **1123b** and the outer surface **1114b** in the radial direction, and there is a gap between the inner surface **1123b** and the outer peripheral surface **1142** of the spacer member **1140** in the radial direction, so that the second joint component **1120** can rotate relative to the spacer member **1140** around the axial direction A without applying a rotational torque to spacer member **1140**. Specifically, at least one of the cross sections of the inner surface **1123b** and the outer peripheral surface **1142** taken along a direction perpendicular to the axial direction A has a circular shape. In this embodiment, both the cross sections of the inner surface **1123b** and the outer peripheral surface **1142** taken along a direction perpendicular to the axial direction A have a circular shape.

[0109] Referring to FIG. 6, FIG. 8, and FIG. 9, a part of the head portion **1131** is disposed opposite to the inner surface **1123b** of the second joint component **1120** in the radial direction. However, due to the gap between the head portion **1131** and the inner surface **1123b**, the transfer of the rotational torque from the second joint component **1120** to the connecting member **1130** can be effectively prevented.

[0110] In other embodiments, the second joint component **1120** may not be provided with the second recess **1123**, but the spacer member **1140** directly abuts against the surface of the second joint component **1120** in the direction parallel to the axial direction A. Moreover, it is not necessary to make the second joint component **1120** opposite to the spacer member **1140** in the radial direction, i.e., it is not necessary to provide the inner surface **1123b**, and thus there is no case in which the inner surface **1123b** is disposed opposite to head portion **1131** in the radial direction.

[0111] In other embodiments, in order to ensure that the spacer member **1140** is unable to rotate relative to the first joint component **1110**, the spacer member **1140** may be fixed directly to the first joint component **1110**, or the spacer member **1140** and the first joint component **1110** may be integrally formed.

[0112] The first joint component **1110**, the second joint component **1120**, the spacer member **1140**, and the connecting member **1130** are assembled as follows:

[0113] Firstly, the second joint component **1120** is sleeved outside of the shaft portion **1112** along the direction parallel to the axial direction A to position the second recess **1123** radially outward of the first recess **1114**, and to position the second bottom surface **1123a** of the second recess **1123** flush with the first bottom surface **1114a** of the first recess **1114**.

[0114] Secondly, the spacer member **1140** is received in the first recess **1114** and the second recess **1123** along the direction parallel to the axial direction A, such that the spacer member **1140** abuts against the first bottom surface **1114a** and the second bottom surface **1123a** in the direction parallel to the axial direction A, and the spacer member **1140** is positioned between the outer surface **1114b** of the shaft portion **1112** and the inner surface **1123b** of the second joint component **1120** in the radial direction.

[0115] Finally, the rod portion **1132** of the connecting member **1130** is inserted into the shaft portion **1112** along the axial direction A and threadedly connected to the shaft portion **1112**, and the head portion **1131** is caused to abut against the spacer member **1140** in the direction parallel to the axial direction A.

[0116] The connecting structure **1100** according to the first embodiment of the present disclosure will be further described with reference to FIG. 3A, FIG. 3B, and FIG. 8. The locking member

1150 is disposed between the first joint component **1110** and the second joint component **1120** in the direction parallel to the axial direction A, and is capable moving between a locking position and an unlocking position along the direction parallel to the axial direction A. In the locking position, the locking member **1150** prevents the first joint component **1110** and the second joint component **1120** from rotating relative to each other. In the unlocking position, the locking member **1150** allows the first joint component **1110** and the second joint component **1120** to rotate relative to each other. The reset member **1170** biases the locking member **1150** toward the locking position. The unlocking member **1160** is provided to be operable externally from the connecting structure **1100** to move the locking member **1150** to the unlocking position.

[0117] Specifically, the locking member **1150** has an annular shape and includes a plurality of locking portions **1151** protruded radially outwardly. The locking member **1150** is sleeved outside of the shaft portion **1112** of the first joint component **1110**. The locking portions **1151** are insertable into guide grooves **1113** around the shaft portion **1112** of the first joint component **1110**. More specifically, the locking portions **1151** are in form of toothed protrusions distributed along the outer periphery of the locking member **1150**, such that the locking member **1150** is overall gear-shaped. The guide grooves **1113** are in form of axial grooves corresponding to the locking portions **1151** to guide the movement of the locking member **1150** along the direction parallel to the axial direction A. The locking position of the locking member **1150** is a position where the locking member **1150** is engaged with the second joint component **1120** in the direction parallel to the axial direction A. The unlocking position of the locking member **1150** is a position where the locking member **1150** is disengaged from the second joint component **1120** in the direction parallel to the axial direction A. It should be understood that the locking member **1150** always remains engaged with the guide grooves **1113** of the first joint component **1110** and is unable to rotate relative to the first joint component **1110**.

[0118] The reset member **1170** is a compression spring in this embodiment and is provided between the first disk-shaped portion **1111** of the first joint component **1110** and the locking member **1150**, thus biasing the locking member **1150** toward the second joint component **1120**. Specifically, the reset member **1170** is sleeved outside of the shaft portion **1112**. In other embodiments, the reset member **1170** may be a component made of an elastomeric material or the like, as long as it can apply a biasing force to the locking member **1150**.

[0119] The second disk-shaped portion **1121** of the second joint component **1120** is provided with a plurality of locking grooves **1122** distributed circumferentially. When the locking member **1150** is in the locking position, the locking portions **1151** is engaged with the respective locking grooves **1122** of the second joint component **1120**. Since the locking member **1150** is unable to rotate relative to the first joint component **1110**, the relative rotation between the first joint component **1110** and the second joint component **1120** is locked when the locking portions **1151** of the locking member **1150** engage into the respective locking grooves **1122** of the second joint component **1120**. Since the locking portions **1151** are evenly distributed circumferentially and the locking grooves **1122** are correspondingly distributed circumferentially, the locking member **1150** is capable of locking the second joint component **1120** at different rotation angles.

[0120] The unlocking member **1160** is provided at an outer side of the first joint component **1110** and the second joint component **1120** in the axial direction A to cover the connecting member **1130**, specifically to cover the head portion **1131**. An outer side of the unlocking member **1160** (i.e., a side of the unlocking member **1160** facing away from the locking member **1150**) has a decorative surface **1161** to provide the connecting structure **1100** a neat and tidy appearance. The locking member **1150** is positioned between the reset member **1170** and the unlocking member **1160** in the direction parallel to the axial direction A and is movably connected to the second joint component **1120**. Specifically, the unlocking member **1160** extends through the second joint component **1120** to abut against the locking member **1150**, such that the unlocking member **1160** can be pressed inwardly (i.e., pressed toward the locking member **1150**) in the direction parallel to the axial

direction A to drive the locking member **1150** to move to the unlocking position. For example, the unlocking member **1160** is provided with at least one pressing portion **1162** extending toward the locking member **1150**. The pressing portion **1162** extends through the second joint component **1120** and moves relative to the second joint component **1120** in the direction parallel to the axial direction A to abut against the locking member **1150**. The pressing portion **1162** is provided with a barb to prevent the detachment of the pressing portion **1162** from the second joint component **1120** when the pressing portion **1162** is subjected to the resilient force of the reset member **1170**. As such, the user can easily operate the locking member **1150** via the unlocking member **1160**.

[0121] In this embodiment, the specific structures of the locking member **1150**, the unlocking member **1160**, and the reset member **1170** advantageously utilize the space between the first joint component **1110** and the second joint component **1120**, and do not conflict with the connecting member **1130**, the spacer member **1140**, and the shaft portion **1112**. It should be understood, however, that the locking member **1150**, the unlocking member **1160**, and the reset member **1170** may be provided in other forms known in the art.

[0122] Considering that the connecting member **1130** is provided in the first joint component **1110** and the connecting member **1130** is used to prevent the detachment of the second joint component **1120** from the first joint component **1110** along the axial direction A, the reinforcing member **1180** is provided inside the first joint component **1110** to reinforce the first joint component **1110**. For example, the first joint component **1110** is made of plastic and the reinforcing member **1180** is made of metal, thereby reducing the overall weight of the connecting structure **1100** while ensuring the necessary strength. In other embodiments, the reinforcing member **1180** may be provided at the outer side of the first joint component **1110** (i.e., the side of the first joint component **1110** facing away from the second joint component **1120**).

[0123] The connecting structure **1100** according to the first embodiment of the present disclosure will be further described with reference to FIG. 7.

[0124] FIG. 7 clearly illustrates the first disk-shaped portion **1111**, the shaft portion **1112**, and the guiding grooves **1113** of the first joint component **1110**. As previously described, the locking member **1150** moves between the locking position and the unlocking position relative to the first joint component **1110** along the direction parallel to the axial direction A, and always remains engaged with the guiding grooves **1113**.

[0125] The connecting structure **1100** according to the first embodiment of the present disclosure will be further described with reference to FIG. 8.

[0126] FIG. 8 illustrates the locking member **1150** in the locking position, where the locking member **1150** is partially within the guiding grooves **1113** of the first joint component **1110** and partially within the locking grooves **1122** of the second joint component **1120**, so that the relative rotation between the first joint component **1110** and the second joint component **1120** is locked. The user may press the locking member **1150** via the unlocking member **1160** to move the locking member **1150** toward the unlocking position (i.e., toward the right side of FIG. 8) and out of the locking grooves **1122** of the second joint component **1120**, thereby allowing the relative rotation between the first joint component **1110** and the second joint component **1120**.

[0127] In summary, the first embodiment of the present disclosure provides a connecting structure having an anti-loosening structure. Specifically, the connecting member connects the second joint component to the first joint component along the axial direction, but there is no direct contact between the connecting member and the second joint component in both the axial and radial directions, avoiding the contact between the connecting member and the second joint component which may transfer a rotational torque from the second joint component to the connecting member and thus cause the detachment of first joint component from the second joint component along the axial direction. It should be understood that although the connecting structure of the present disclosure is described in connection with a stroller, the connecting structure may be applied to other devices.

Second Embodiment

[0128] A foldable stroller is typically equipped with connecting joints on both the left and right sides. The connecting joints at the left and right sides can be unlocked using a flexible webbing, allowing a seat and a frame of the stroller to fold synchronously. In the unlocking process by lifting and pulling the webbing, the two ends of the flexible webbing may experience unbalanced forces and swaying, especially if the stroller or the frame thereof is relatively heavy, which may lead to failure in unlocking the connecting joints at the left and right sides of the frame synchronously. In addition, in the process of unlocking and folding the stroller via the flexible webbing, if the position of lifting and pulling the flexible webbing deviates from the substantially middle of the flexible webbing, the stroller or the frame of the stroller may tilt, and the flexible webbing may fail to transmit sufficient driving force to the connecting joints, which requires the user to lift the stroller and further shake it forcefully to unlock the connecting joints. This process demands more effort from the user. In other words, using the flexible webbing to unlock and fold the stroller or the frame of the stroller may not be an effortless task.

[0129] In view of the above issues, the second embodiment of the present disclosure provides a child carrier and a child carrying device thereof, as shown in FIG. 10 to FIG. 20.

[0130] Referring to FIG. 10, a child carrier **2000** provided according to the second embodiment of the present disclosure is shown. The child carrier **2000** includes a supporting device **2300** and a child carrying device **2400** provided according to the second embodiment of the present disclosure. The child carrying device **2400** is disposed on the supporting device **2300**. The child carrying device **2400** includes a folding joint **2200**. The supporting device **2300** includes a connecting joint **2100**. The folding joint **2200** is connected to the connecting joint **2100** such that the child carrying device **2400** is connected to the supporting device **2300**. Exemplarily, the folding joint **2200** is removably connected to the connecting joint **2100**, such that the child carrying device **2400** is removably connected to the supporting device **2300**. In the second embodiment of the present disclosure, the child carrier **2000** is described with a stroller as an example. The supporting device **2300** is specifically a frame of the stroller. The child carrying device **2400** may be a foldable seat, a folding frame for use in combination with a baby cradle, or the like. Specifically, the child carrying device **2400** is described with a foldable seat as an example. In alternative embodiments, the form of the child carrier **2000** is not limited to a stroller, but may be, for example, a child bed frame or other products. The forms of the supporting device **2300** and the child carrying device **2400** may vary depending on the form of the child carrier **2000**.

[0131] Referring to FIG. 10, in some embodiments, the child carrier **2000** has a substantially left-right symmetrical structure. The supporting device **2300** includes, for example, a handlebar **231**, a front leg frame **232**, a rear leg frame **233**, and the connecting joint **2100**. The handlebar **231** and the front leg frame **232** are connected to each other by one connecting joint **2100** at each of right and left sides of the supporting device **2300**. The handlebar **231** is pivotally connected to the rear leg frame **233**. Referring also to FIG. 15, more specifically, the connecting joint **2100** includes a first connecting base **211** and a second connecting base **212**. The first connecting base **211** and the second connecting base **212** are pivotally connected to each other via a pivot shaft **2101** (see FIG. 15) such that they can rotate relative to each other. The pivot shaft **2101** defines a pivot axis R1 of the connecting joint **2100** (see, for example, FIG. 15).

[0132] Referring again to FIG. 10, the child carrying device **2400** includes a seat tube **241**, a backrest tube **242**, an operation assembly **2500**, and the folding joint **2200**. The seat tube **241** and the backrest tube **242** are connected to each other by one folding joint **2200** at each of right and left sides of the child carrying device **2400**. The folding joint **2200** includes a mounting base **221**, a first pivoting base **222**, and a second pivoting base **223**. The mounting base **221**, the first pivoting base **222**, and the second pivoting base **223** are pivotally connected to each other via the first pivot shaft **2201** such that they can rotate relative to each other. The first pivot shaft **2201** defines a pivot axis R2 of the folding joint **2200**. The mounting base **221** is configured to be removably mounted

to the second connecting base **212** of the connecting joint **2100** such that the child carrying device **2400** is removably mounted to the supporting device **2300**. The second pivoting base **223** has a connecting sleeve **22301**. The backrest tube **242** is sleeved into the connecting sleeve **22301**. [0133] It should be noted that, unless otherwise expressly specified and defined, in the second embodiment of the present disclosure, the orientation terms such as “front”, “rear”, “left”, and “right” are based on the orientations of a child riding in the child carrier. In the drawings, the ‘front’ and ‘rear’ directions are indicated by arrows P and B, respectively, while the ‘left’ and ‘right’ directions are indicated by arrows L and R, respectively. These orientation terms are used only to make the description of the second embodiment of the present disclosure clearer and are not intended to unduly limit the scope of protection of the present disclosure.

[0134] Referring to FIG. 15 and FIG. 16, an exemplary structure of the second connecting base **212** of the connecting joint **2100** and the mounting base **221** of the folding joint **2200** are illustrated. Referring also to FIG. 21 and FIG. 22, the first connecting base **211** is positioned at a first side **212a** of the second connecting base **212**, which may also be referred to as an outer side of the second connecting base **212**. A locking hole **2120** is provided on a second side **212b** of the second connecting base **212**, which may also be referred to as an inner side of the second connecting base **212**. An engagement table **2121** that is protruded outwardly is also provided on the second side **212b** of the second connecting base **212**. The engagement table **2121** is positioned under the locking hole **2120**. An end face of the engagement table **2121** (i.e., an end face of the second connecting base **212** on the second side **212b**) is provided with a mounting recess **21210**. Referring also to FIG. 20, the mounting base **221** is provided with an engagement groove **2210** and a through-hole **22108** on a side thereof facing the second connecting base **212**. The mounting base **221** is mounted with a locking member **22101**, an unlocking member **22102**, and a resilient reset member **22103**. The locking member **22101** is, for example, a locking pin. The resilient reset member **22103** drives the locking member **22101** to extend out of the through-hole **22108**. The unlocking member **22102** is operably connected to the locking member **22101** to drive the locking member **22101** to retract into the through-hole **22108**. When the mounting base **221** is mounted to the second connecting base **212**, the engagement groove **2210** of the mounting base **221** is engaged with the engagement table **2121** of the second connecting base **212**, and the locking member **22101** is inserted into the locking hole **2120**. In order to remove the mounting base **221**, the unlocking member **22102** can be pressed to drive the locking member **22101** to withdraw from the locking hole **2120**, and then the engagement groove **2210** of the mounting base **221** can be disengaged from the engagement table **2121** of the second connecting base **212**.

[0135] Referring to FIG. 15 and FIG. 16, in some embodiments, the connecting joint **2100** further includes a detent member **213**, a first release member **214**, and a drive member **216**. The detent member **213** is movably disposed between the first connecting base **211** and the second connecting base **212**, and has a detent position and a release position. The first release member **214** is operably connected to the detent member **213** via the drive member **216**, such that the first release member **214** drives the detent member **213** indirectly via the drive member **216** to move from the detent position to the release position. In some embodiments, a chamber **210** is formed between the first connecting base **211** and the second connecting base **212**. The detent member **213** may be a detent gear. The detent member **213** may be disposed in the chamber **210** and axially movable in the chamber **210**. More specifically, referring also to FIG. 21 and FIG. 22, the pivot shaft **2101** extends through a center hole **21100** of the first connecting base **211**, a center hole **21300** of the detent member **213**, and a center hole **21200** of the second connecting base **212**. Referring to FIG. 21 and FIG. 22, in some embodiments, the first connecting base **211** has a first inner ring gear **2113**, and the second connecting base **212** has a second inner ring gear **2123**. Referring to FIG. 15, when the detent member **213** is in the detent position, the detent member **213** is engaged with both the first inner ring gear **2113** and the second inner ring gear **2123** to fix the relative position between the first connecting base **211** and the second connecting base **212**. Referring to FIG. 16, when the

detent member **213** is in the release position, the detent member **213** is moved into the first connecting base **211** along the pivot shaft **2101** to disengage from the second inner ring gear **2123** of the second connecting base **212**, so that the first connecting base **211** and the second connecting base **212** is allowed to rotate relative to each other. In some alternative embodiments, the detent member **213** may be implemented in other ways, such as in a pin structure, which is not limited in the present disclosure.

[0136] Referring to FIG. **21** and FIG. **22**, in some embodiments, a resilient reset member **215** may also be disposed in the chamber **210**. The resilient reset member **215** is configured to apply a force to the detent member **213** to reset the detent member **213** to the detent position. In some embodiments, the resilient reset member **215** is, for example, a spring mounted between the first connecting base **211** and the detent member **213**. In some alternative embodiments, the implementation of the resilient reset member **215** may vary depending on the implementation of the detent member **213**.

[0137] Refer again to FIG. **21** and FIG. **22**, the detent member **213** has at least one limiting tooth **2131** with a tooth width greater than that of other teeth of the detent member **213**. Correspondingly, the first inner gear ring **2113** is provided with at least one first limiting tooth groove **21131**, while the second inner gear ring **2123** is provided with at least one second limiting tooth groove **21231**. The number of each of the first limiting tooth groove **21131** and the second limiting tooth groove **21231** corresponds to the number of the limiting tooth **2131**. The at least one limiting tooth **2131** is adapted to engage with each of the at least one first limiting tooth groove **21131** and the at least one second limiting tooth groove **21231**. Exemplarily, there are two limiting teeth **2131** arranged axisymmetrically on the outer periphery of the detent member **213**. When the supporting device **2300** is in the unfolded state, the first limiting tooth grooves **21131** are aligned with the respective second limiting tooth grooves **21231**, and two ends of each of the limiting teeth **2131** are adapted to be engaged with corresponding ones of the first limiting tooth grooves **21131** and the second limiting tooth grooves **21231**, respectively. When connecting joint **2100** is in the locked state, the first connecting base **211** is unable to rotate relative to the second connecting base **212**. When the supporting device **2300** switches between the unfolded state and the folded state, the detent member **213** is positioned in the first connecting base **211** and disengaged from the second connecting base **212**, the first connecting base **211** rotates relative to the second connecting base **212**, the second limiting tooth grooves **21231** are misaligned with the first limiting tooth grooves **21131** (and the limiting teeth **2131**), the end of the second inner ring gear **2123** blocks the limiting teeth **2131** and resists the force applied to the detent member **213** by the resilient reset member **215** to restrict the detent member **213** from moving to the detent position, thereby restricting the detent member **213** from being engaged with the second inner ring gear **2123** of the second connecting base **212**, consequently, the detent member **213** is limited to the release position, allowing the first connecting base **211** to continue to rotate relative to the second connecting base **212** until the supporting device **2300** is switched to the folded state. In this way, the support device **2100** can be switched between the unfolded state and the folded state without requiring to continuously push the first release member **214**, which is convenient for operation.

[0138] Referring to FIG. **15** and FIG. **16**, the drive member **216** is interposed between the first release member **214** and the detent member **213**. The first release member **214** drives the detent member **213** indirectly via the drive member **216**. In some embodiments, the drive member **216** includes, for example, a third push portion **2161** and a third abutment portion **2162**. The third abutment portion **2162** is, for example, disk-shaped. The third push portion **2161** includes, for example, at least one leg portion extending from the third abutment portion **2162**. The third abutment portion **2162** is positioned in the mounting recess **21210** and abuts against the first push portion **2141** of the first release member **214**. A wall of the second connecting base **212** between the mounting recess **21210** and the chamber **210** is provided with at least one through-hole **21208**. The third push portion **2161** extends through the through-hole **21208** to abut against the detent

member **213**. The third push portion **2161** is slidably fitted in the through-hole **21208** to guide the drive member **216** to move in the axial direction of the pivot shaft **2101**. By providing the drive member **216** that moves in the axial direction, the design difficulty of the first release member **214** can be reduced, as long as the first release member **214** can push the drive member **216** via the first push portion **2141**. In some alternative embodiments, the drive member **216** may be omitted, the first push portion **2141** of the first release member **214** includes, for example, at least one protruded structure such as a leg or a cylinder (not shown), and the protruded structure of the first push portion **2141** extends through the at least one through-hole **21208** of the second connecting base **212** to directly abut against the detent member **213**. This facilitates a reduction in the number of parts and ease of assembly.

[0139] Referring to FIG. 15 and FIG. 16, in some embodiments, the first release member **214** is mounted to the second connecting base **212**. Specifically, the first release member **214** is pivotally connected to the second connecting base **212** by a first shaft **21211** perpendicular to the pivot shaft **2101**. The first release member **214** includes a first push portion **2141** and a first abutment portion **2142**. The first push portion **2141** extends, for example, from the first abutment portion **2142**. When the first release member **214** is subjected to an external driving force (e.g., an external force directly applied to the first abutment portion **2142**, or an external force indirectly transmitted to the first abutment portion **2142** via the folding joint **2200**, etc.), the first release member **214** rotates around the first shaft **21211**, while the first push portion **2141** of the first release member **214** directly or indirectly pushes the detent member **213** to move from the detent position to the release position. It is to be understood that the external driving force refers to a force from outside of the connecting joint **2100** where the first release member **214** is located. Referring to FIG. 21 and FIG. 22, the first release member **214** is disposed at the second side **212b** of the second connecting base **212**, and the second side **212b** of the second connecting base **212** is configured to connect to the mounting base **221**. In some embodiments, the first release member **214** is disposed in the mounting recess **21210**. When the child carrying device **2400** is mounted to the supporting device **2300**, the first release member **214** is covered by the mounting base **221**, which prevents the first release member **214** from being accidentally touched. More specifically, the accidental folding of the supporting device **2300** due to accidental touching of the first release member **214** when the child carrying device **2400** is in the unfolded state and supports a child can be prevented.

[0140] Referring to FIG. 15 and FIG. 16, in some embodiments, the connecting joint **2100** further includes a second release member **217** operably connected to the first release member **214**. The second release member **217** is positioned in the mounting recess **21210** and is pivotally connected to the second connecting base **212** by a second shaft **21212**. The second shaft **21212** is parallel to the first shaft **21211**. A first end of the second release member **217** is engaged with the first release member **214**. A second end of the second release member **217** is provided with a second abutment portion **2172**. When the second release member **217** is subjected to no external driving force, the second abutment portion **2172** protrudes out of the second connecting base **212**. When the second abutment portion **2172** of the second release member **217** is subjected to an external driving force, the first release member **214** is driven to rotate due to the engagement of the first end of the second release member **217** with the first release member **214**. That is, the first release member **214** can be indirectly driven to rotate by pressing the second abutment portion **2172** of the second release member **217**. It is to be understood that an external driving force is a force that comes from outside of the connecting joint **2100** in which the second release member **217** is located. The first release member **214** has a first connecting portion **2143** sleeved onto the first shaft **21211**. The first connecting portion **2143** has a first tooth structure (tooth) **21431**. The second release member **217** has a second connecting portion **2171** sleeved onto the second shaft **21212**. The second connecting portion **2171** has a second tooth structure (tooth) **21711**. The first tooth structure **21431** of the first connecting portion **2143** is engaged with the second tooth structure **21711** of the second connecting portion **2171**. This gear drive mechanism between the first release member **214** and the second

release member **217** can, on the one hand, increase the pushing stroke of the first release member **214**, and on the other hand, reduce the pushing force of the second release member **217** and increase the pushing stability of the second release member **217**, thereby increasing the stability and effectiveness of the locking process of the connecting joint **2100**, and realizing the rapid folding of the connecting joint **2100**. In some alternative embodiments, the first release member **214** and the second release member **217** may be engaged with other or movably connected to each other by other suitable structures.

[0141] Referring to FIG. **15**, FIG. **16**, FIG. **21**, and FIG. **22**, in some embodiments, the connecting joint **2100** may further include a fixing bracket **218**. The fixing bracket **218** includes, for example, two side walls **2182** opposite to each other and a plate **2181** connected between the two side walls **2182**. The plate **2181** and the two side walls **2182** define a cavity **2180**. The two side walls **2182** are each provided with a first mounting hole **21821** and a second mounting hole **21822**. The first shaft **21211** is mounted through the first mounting holes **21821** of the two side walls **2182**. The second shaft **21212** is mounted through the second mounting holes **21822** of the two side walls **2182**. The first connecting portion **2143** and the second connecting portion **2171** are accommodated in the cavity **2180**. The fixing bracket **218** is mounted in the mounting recess **21210**. The opening of the mounting recess **21210** is partially closed by the plate **2181** to prevent the external parts from interfering with the engagement of the first connecting portion **2143** with the second connecting portion **2171**.

[0142] Referring to FIG. **15** and FIG. **16**, in some embodiments, the first release member **214** rotates around the first shaft **21211** when it is driven by the second release member **217** or the folding joint **2200** or when it is directly pressed by the user. The first release member **214** in rotation can drive the drive member **216** to release the detent member **213** such that the detent member **213** can move from the detent position to the release position. It should be appreciated that, upon removal of the driving force applied to the first release member **214**, the detent member **213** is driven and reset by the first resilient reset member **215** from the release position to the detent position, and accordingly, the drive member **216** is driven and reset by the detent member **213**, while the first release member **214** is driven and reset by the drive member **216**. In some alternative embodiments, a torsion spring for resetting the first release member **214** may be mounted on the first shaft **21211**.

[0143] Referring to FIG. **11** to FIG. **16**, the schematic structures of the child carrying device **2400**, the folding joint **2200** of the child carrying device **2400**, and the operation assembly **2500** in FIG. **10** is shown.

[0144] Referring to FIG. **15** and FIG. **16**, the operation assembly **2500** is operatively connected to the folding joint **2200** and can drive the folding joint **2200** to switch from the locked state to the unlocked state (i.e., the operation assembly **2500** can unlock the folding joint **2200**). The folding joint **2200** includes a push member **227**, a folding locking member **228**, the mounting base **221**, the first pivoting base **222**, and the second pivoting base **223**. Referring also to FIG. **19** and FIG. **20**, the first pivot shaft **2201** extends through a center hole **22100** of the mounting base **221**, a center hole **22200** of the first pivoting base **222**, and a center hole **22300** of the second pivoting base **223**, such that the mounting base **221**, the first pivoting base **222**, and the second pivoting base **223** are pivotably connected to each other via the first pivot shaft **2201** and thus can rotate relative to each other. The first pivot shaft **2201** defines a pivot shaft R2 of the folding joint **2200**. The first pivoting base **222** is sandwiched between the mounting base **221** and the second pivoting base **223**, that is, the first pivoting base **222** is disposed at the side of the second pivoting base **223** facing the mounting base **221**. The folding locking member **228** is disposed between the first pivoting base **222** and the mounting base **221** and is adapted to lock the relative rotation between the first pivoting base **222** and the mounting base **221**. The push member **227** is disposed between the operation assembly **2500** and the second pivoting base **223**, that is, the push member **227** is disposed at the side of the second pivoting base **223** facing away from the mounting base **221**. The

push member **227** is adapted to cooperate with the operation assembly **2500** and is indirectly or directly drivingly connected to the folding locking member **228**. The operation assembly **2500** is adapted to drive the folding locking member **228** indirectly via the push member **227** or directly to unlock the relative rotation between the first pivoting base **222** and the mounting base **221**.

[0145] Referring to FIG. **15**, FIG. **16**, and FIG. **18**, in some embodiments, a chamber **220** is formed between the mounting base **221** and the first pivoting base **222**. The folding locking member **228** is received in the chamber **220**. The folding locking member **228** may be a toothed wheel with a center hole **22800** (see FIG. **20**) sleeved onto the first pivot shaft **2201**. The folding locking member **228** is disposed in the chamber **220** and axially movable in the chamber **220**. The folding locking member **228** has a locking position and an unlocking position. When the folding locking member **228** is in the locking position, the folding locking member **228** engages with both the mounting base **221** and the first pivoting base **222** to fix the relative position of the mounting base **221** and the first pivoting base **222**, such that the folding joint **2200** is in the locked state. When the folding locking member **228** is in the unlocking position, the folding locking member **228** is moved into the mounting base **221** along the first pivot shaft **2201** and disengaged from the first pivoting base **222** to allow the relative rotation of the mounting base **221** and the first pivoting base **222**, such that the folding joint **2200** is in the unlocked state. In some embodiments, a resilient reset member **226** is further mounted in the chamber **220**. The resilient reset member **226** is configured to drive the folding locking member **228** to move to the locking position. The resilient reset member **226** is, for example, a spring that is interposed between the mounting base **2221** and the folding locking member **28**.

[0146] Referring to FIG. **19** and FIG. **20**, the folding locking member **228** is provided with an outer ring gear **2283**. The outer ring gear **2283** includes at least one limiting tooth **22831** having a tooth width greater than those of other teeth of the outer ring gear **2283**. Accordingly, the first pivoting base **222** is provided with a first inner ring gear **2224** formed with at least one limiting tooth groove **22241**, and the mounting base **221** is provided with a second inner ring gear **2214** formed with at least one second limiting tooth groove **22141**. When the folding locking member **228** is in the locking position, the limiting tooth **22831** is adapted to engage with each of the first limiting tooth groove **22241** and the second limiting tooth groove **22141**. Exemplarily, two limiting teeth **22831** are provided, while two first limiting tooth grooves **22241** and two second limiting tooth grooves **22141** are provided. Referring also to FIG. **15**, when the child carrying device **2400** is in the unfolded state (not shown), the first limiting tooth grooves **22241** are aligned with respective second limiting tooth grooves **22141**, and each of the limiting teeth **22831** of the folding locking member **228** is engaged with corresponding ones of the first limiting tooth grooves **22241** and of the second limiting tooth grooves **22141** (i.e., in the direction of the pivot axis **R2**, a part of each of the limiting teeth **22831** is engaged with the corresponding one of the first limiting tooth grooves **22241**, and another part of each of the limiting teeth **22831** is engaged with the corresponding one of the second limiting tooth grooves **22141**) to restrict the relative rotation between the first pivoting base **222** and the mounting base **221**. Referring also to FIG. **16**, when the child carrying device **2400** switches between the unfolded state and the folded state, the folding locking member **228** is positioned within the mounting base **221** and disengaged from the first pivoting base **222**, the first pivoting base **222** rotates relative to the mounting base **221**, the second limiting tooth grooves **22141** are misaligned with the first limiting tooth grooves **22241**, and the end of the first inner gear ring **2224** blocks the limiting teeth **22831** and resists the force of the resilient reset member **226** (see FIG. **18**) applied to the folding locking member **228** to restrict the folding locking member **228** from moving to the locking position, thereby restricting the folding locking member **228** from engaging with the first inner ring gear **2224** of the first pivoting base **222**, consequently, the folding locking member **228** is limited to the unlocking position, and the first pivoting base **222** and the mounting base **221** can continue to pivot relative to each other, enabling the child carrying device **2400** to switch to the folded state. In this way, the folding locking member **228** can be held

in the unlocking position to allow the first pivoting base **222** and the second pivoting base **223** to pivot downwardly under gravity, without the need to continually apply an external force to the operation assembly **2500**, which makes the operation convenient.

[0147] Referring to FIG. **15**, FIG. **16**, and FIG. **18**, the folding locking member **228** is provided with at least one protruding finger **2281**. A wall of the mounting base **221** between the engagement groove **2210** and the chamber **220** is provided with at least one through-hole **22107**. The at least one protruding finger **2281** is adapted to extend through the at least one through-hole **22107** to push the first release member **214** to rotate around the first shaft **21211**. Exemplarily, the folding locking member **228** is provided with a plurality of protruding fingers **2281**, such as two protruding fingers **2281**. The protruding fingers **2281** are uniformly arranged in a circumferential direction of the folding locking member **228** and collectively guide the axial movement of the folding locking member **228**. The protruding fingers **2281** are slidably fitted in the respective through-holes **22107**. It should be appreciated that the sliding direction of the individual protruding fingers **2281** is parallel to the first pivot shaft **2201**. As the folding locking member **228** switches between the locking position and the unlocking position, the at least one protruding finger **2281** telescopes with respect to the at least one through-hole **22107**. Referring to FIG. **15** to FIG. **16**, during the movement of the folding locking member **228** from the locking position to the unlocking position, the at least one protruding finger **2281** protrudes out of the at least one through-hole **22107** and directly pushes against the first abutment portion **2142** of the first releasing member **214**, causing the first releasing member **214** to rotate around the first shaft **21211** to indirectly or directly (via or not via the drive member **216**) drive the first detent member **213** to move from the detent position to the release position. That is, the connecting joint **2100** can be unlocked by the folding joint **2200** during the unlocking of the folding joint **2200** to allow for the folding of the supporting device **2300**.

[0148] Referring to FIG. **18** to FIG. **20**, in some embodiments, a movable piece **229** is also mounted in the chamber **220**. The movable piece **229** has a center hole **2290** sleeved onto the first pivot shaft **2201**. The movable piece **229** is positioned between the folding locking member **228** and the first pivoting base **222**. The movable piece **229** is provided with at least one abutment post **2291** extending away from the folding locking member **228**. The first pivoting base **222** has at least one through-hole **22203**. The at least one abutment post **2291** is slidably fitted in the at least one through-hole **22203**. The folding locking member **228** is drivably connected to the push member **227** via the movable piece **229** and the at least one abutment post **2291**.

[0149] Referring to FIG. **18** to FIG. **20**, in some embodiments, the center hole **22700** of the push member **227** is sleeved onto the first pivot shaft **2201** (see FIG. **15**). The push member **227** has at least one push post **2271**. The second pivoting base **223** has at least one through-hole **22303**. The at least one push post **2271** is slidably fitted in the at least one through-hole **22303**. The at least one push post **2271** is configured to push the at least one abutment post **2291** to cause the movable piece **229** to push the folding locking member **228** to move from the locking position to the unlocking position. In some embodiments, the folding joint **2200** may also be provided with a resilient reset member **2232** that is configured to drive the push member **227** to move away from the movable piece **229**. The resilient reset member **2232** is, for example, a spring interposed between the push member **227** and the second pivoting base **223**.

[0150] Referring to FIG. **11** to FIG. **14**, in some embodiments, the operation assembly **2500** includes an operation member **251** and a folding unlocking member **252**. The folding unlocking member **252** is operably connected to the folding joint **2200** to unlock the folding joint **2200**. The folding unlocking member **252** is pivotally connected to the first pivot shaft **2201** and is capable of rotating relative to the second pivoting base **223** to drive the push member **227** to move in the axial direction, such that the push post **2271** of the push member **227** pushes the abutment post **2291** of the movable piece **2221** to unlock the folding joint **2200**. Referring to FIG. **11** and FIG. **13**, the folding unlocking member **252** has a first position and a second position. When the folding

unlocking member **252** is in the first position, the folding joint **2200** is in the locked state, and the folding locking member **228** is in the locking position and restricts the rotation of the first pivoting base **222** relative to the mounting base **221**. Referring to FIG. **12**, when the folding unlocking member **252** rotates from the first position to the second position, the folding unlocking member **252** rotates relative to the second pivoting base **223** and unlock the folding joint **2200** indirectly via the push member **227** (i.e., drive the folding locking member **228** from the locking position to the unlocking position indirectly via the push member **227**), thereby allowing the rotation of the first pivoting base **222** relative to the mounting base **221**. There are, for example, two folding unlocking members **252** connected to the folding joints **2200** at right and left sides, respectively. There is, for example, one operation member **251** connected to each of the two folding unlocking members **252**. [0151] For ease of description, one of the folding unlocking members **252** and one of the folding joints **2200** connected thereto are illustrated below as an example.

[0152] Referring to FIG. **19** and FIG. **20**, in some embodiments, the folding unlocking member **252** is provided with a first cooperation portion thereon. The push member **227** is provided with a second cooperation portion thereon. The first cooperation portion is configured to cooperate with the second cooperation portion in a concave-convex fitting manner and to be slidable relative to the second cooperation portion along the surface of the second cooperation portion. The first cooperation portion is provided with at least one first convex portion **2521** and at least one first concave portion **2529**. Each first convex portion **2521** at least partially define a wall of the first concave portion **2529** adjacent thereto. There is a smooth transition between a highest point of each first convex portion **2521** and a lowest point of the adjacent first concave portion **2529**. The at least one first convex portion **2521** includes, for example, at least one tooth structure. Exemplarily, there are, for example, at least two first convex portions **2521**. Any two adjacent first convex portions **2521** define one first concave portion **2529** therebetween. The second cooperation portion is provided with at least one second convex portion **2272** and at least one second concave portion **2273**. The at least one second convex portion **2272** is capable of cooperating with the at least one first concave portion **2529** in a concave-convex fitting manner. The at least one second concave portion **2273** is capable of cooperating with the at least one convex portion **2521** in a concave-convex fitting manner. Each second convex portion **2272** at least partially define a wall of the concave portion **2273** adjacent thereto. There is a smooth transition between a highest point of each second convex portion **2272** and a lowest point of the adjacent second concave portion **2273**. This facilitates the withdrawal of each convex portion **2521** from the corresponding second concave portion **2273**. Specifically, the at least one second convex portion **2272** includes, for example, at least one tooth structure.

[0153] When the folding unlocking member **252** is in the first position, the at least one first convex portion **2521** is in concave-convex fit with the at least one second concave portion **2273**, and the at least one second convex portion **2272** is in concave-convex fit with the at least one first concave portion **2529**, that is, the first cooperation portion is in concave-convex fit with the second cooperation portion. As the folding unlocking member **252** rotates from the first position to the second position, the first cooperation portion slides along the surface of the second cooperation portion to cause the at least one first convex portion **2521** to abut against the at least one second convex portion **2272**, thereby pushing the push member **227** in a direction away from the folding unlocking member **252** along the first pivot shaft **2201**, such that the push post **2271** of the push member **227** abuts against the abutment post **2291** of the movable piece **229** and drives the folding locking member **228** to move to the unlocking position. It should be appreciated that when the user no longer apply a force to the operation member **251**, the push member **227** has a tendency to move in a direction away from the second pivoting base **223** (i.e., in a direction toward the folding unlocking member **252**) to be reset under the action of the resilient reset member **2232**. When the at least one second convex portion **2272** of the push member **227** is misaligned with the at least one first convex portion **2521** of the folding unlocking member **252** (i.e., when the at least one second

convex portion 2272 of the push member 227 is aligned with at least one first concave portion 2529 of the folding unlocking member 252, and the at least one first convex portion 2521 of the folding unlocking member 252 is aligned with the at least one second concave portion 2273 of the push member 227), the push member 227 is driven and reset by the resilient reset member 2232. It should be understood that when the at least one first convex portion 2521 abuts against the at least one second convex portion 2272, a highest point of at least one of the at least one first convex portion 2521 abuts against a highest point of at least one of the at least one second convex portion 2272.

[0154] Referring again to FIG. 19 and FIG. 20, in some embodiments, the second pivoting base 223 and the folding unlocking member 252 are provided with a rotation limiting mechanism to limit the range of the relative rotation therebetween, i.e., to limit the pivot stroke of the folding unlocking member 252. For example, an arc-shaped groove 22305 may be provided on the second pivoting base 223, and an arc-shaped protrusion 2525 may be provided on the folding unlocking member 252, with the arc-shaped protrusion 2525 positioned in the arc-shaped groove 22305. The ends of the arc-shaped protrusion 2525 can selectively abut against the ends of the arc-shaped groove 22305 to limit the range of the relative rotation between the second pivoting base 223 and the folding unlocking member 252. In some embodiments, the positions of the arc-shaped groove 22305 and the arc-shaped protrusion 2525 on the second pivoting base 223 and the folding unlocking member 252 are interchangeable.

[0155] Referring to FIG. 11 to FIG. 14, the folding unlocking member 252 includes a main body 2522. The main body 2522 is substantially disk-shaped. The first pivot shaft 2201 extends through the main body 2522 at a substantially central position of the main body 2522. Referring also to FIG. 19 and FIG. 20, the arc-shaped protrusion 2525 and the first convex portion 2521 are provided on the side of the main body 2522 facing the push member 227 (also referred to as the inner side of the folding unlocking member 252). The main body 2522 is provided with a limiting recess 2523 thereon. An opening of the limiting recess 2523 is formed substantially at the outer periphery of the main body 2522. The limiting recess 2523 is configured to connect to the operation member 251.

[0156] Referring to FIG. 11 to FIG. 14, in some embodiments, the operation member 251 is a rigid member and is pivotably connected to the folding unlocking member 252 via a second pivot shaft 2501, such that the operation member 251 is capable of rotating relative to the folding unlocking member 252. The second pivot shaft 2501 defines a pivot axis R3 of the operation member 251. The second pivot shaft 2501 is parallel to and offset from the first pivot shaft 2201. The operation member 251 is adapted to drive the folding unlocking member 252 to rotate relative to the second pivoting base 223 (i.e., relative to the folding joint 2200). Specifically, the operation member 251 is adapted to drive the folding unlocking member 252 to rotate relative to the second pivoting base 223 (i.e., relative to the folding joint 2200) via the second pivot shaft 2501 to unlock the folding joint 2200. As compared to the coaxial arrangement of the second pivot shaft 2501 and the first pivot shaft 2201, the offset of the second pivot shaft 2501 from the first pivot shaft 2201 in this embodiment can increase the torque corresponding to the force applied by the user to the operation member 251, thus making the unlocking operation more labor-saving. In addition, by pivotably connecting the rigid operation member 251 to the folding unlocking member 252, the failure of the operation member 251 to transmit sufficient driving force to the folding joint 2200 due to the deformation and skewing of the operation member 251 when it is operated can be avoided. Due to the rigid structure of the operation member 251, the direction of the force applied to the operation member 251 can be well controlled, so that the operation member 251 can more effectively transmit the driving force to the folding joint 2200, and the situation where the folding joints 2200 at two sides are subjected to unevenly distributed loads which causes asynchronized unlocking of the folding joints 2200 can be avoided, thereby achieving the rapid folding of the child carrier 2000 in a more efficient manner with less effort.

[0157] The operation member 251 is pivotally connected to the wall of the limiting recess 2523 via

the second pivot shaft **2501**. Specifically, the wall of the limiting recess **2523** is provided with a pivot hole **2520**. An end of the operation member **251** is inserted into the limiting recess **2523** and provided with a pivot hole **2510**. The second pivot shaft **2501** extends through the pivot holes **2520** and **2510** to pivotally connect the end of the operation member **251** to the wall of the limiting recess **2523**. The limiting recess **2523** has a first side wall **25231** and a second side wall **25232** located at two sides of the second pivot shaft **2501**, respectively. The operation member **251** is adapted to be selectively blocked by the first side wall **25231** and the second side wall **25232**, so that the position and the rotation range of the operation member **251** relative to the folding unlocking member **252** are limited. Specifically, the operation member **251** has a first limited position and a second limited position. When the operation member **251** is in the first limited position, the operation member **251** abuts against the first side wall **25231**. When the operation member **251** is in the second limited position, the operation member **251** abuts against the second side wall **25232**. The folding unlocking member **252** has a first position. When the folding unlocking member **252** is in the first position, the operation member **251** is in the first limited position.

[0158] Refer again to FIG. **11** to FIG. **14**, when the operation member **251** is subjected to a pulling force **F**, the operation member **251** is enabled to rotate relative to the folding unlocking member **252** around the second pivot shaft **2501**, and the operation member **251** can drive the folding unlocking member **252** to rotate from the first position to the second position via the second pivot shaft **2501** to unlock the folding joint **2200**, and can further drive the folding unlocking member **252** to rotate from the second position to a third position. When the folding unlocking member **252** is in the first position, the pulling force **F** applied to the operation member **251** has a vertically upward component. When the folding unlocking member **252** is driven by the operation member **251** to rotate to the second position, the direction of the pulling force **F** applied to the operation member **251** is vertically upward. During the rotation of the folding unlocking member **252** from the second position to the third position, the folding joint **2200** remains in the unlocked state, and the direction of the pulling force **F** applied to the operation member **251** remains in the vertical upward direction.

[0159] Referring to FIG. **10** to FIG. **14**, the operation member **251** is provided with an operation portion **2512** adapted to be operated. The operation portion **2512** is located away from the second pivot shaft **2501**. The operation member **251** is, for example, substantially U-shaped. The operation member **251** includes two connecting portions **2511** and a crossbar connected between the two connecting portions **2511**. The crossbar serves as the operation portion **2512**. The connecting portions **2511** are each, for example, rod-shaped. The two connecting portions **2511** are pivotally connected to the two folding unlocking members **252** via the second pivot shafts **2501**, respectively. As such, the operation of the operation portion **2512** of the operation member **251** can unlock both the folding joints **2200** at right and left sides. Additionally, since the operation member **251** is rigid, the deformation of the operation member **251** in operation is very small (even negligible), so that the child carrier **2000** is not easily tilted even if the force-subjecting position of the operation member **251** is deviated from the center of the operation portion **2512**, and the folding joints **2200** at right and left sides can be unlocked at the same time.

[0160] Referring to FIG. **11**, in some embodiments, when the folding unlocking member **252** is in the first position, the folding joint **2200** is in the locked state, and the operation portion **2512** of the operation member **251** is positioned above the horizontal plane where the first pivot shaft **2201** is located and/or above the horizontal plane where the second pivot shaft **2501** is located. In this way, the operation member **2512** of the operation member **251** can be conveniently lifted and operated. It is to be understood that “the operation portion **2512** of the operation member **251** is positioned above the horizontal plane where the first pivot shaft **2201** is located and/or above the horizontal plane where the second pivot shaft **2501** is located” includes the following different implementation manners: (a) the operation portion **2512** of the operation member **251** is positioned

above the horizontal plane where the first pivot shaft **2201** is located but under the horizontal plane where the second pivot shaft **2501** is located; (b) the operation portion **2512** of the operation member **251** is positioned above the horizontal plane where the second pivot shaft **2501** is located but under the horizontal plane where the first pivot shaft **2201** is located; and (c) the operation portion **2512** of the operation member **251** is positioned above both the horizontal plane where the first pivot shaft **2201** is located and the horizontal plane where the second pivot shaft **2501** is located. In other words, the operation portion **2512** of the operation member **251** is positioned above at least one of the horizontal plane where the first pivot shaft **2201** is located and the horizontal plane where the second pivot shaft **2501** is located. In this embodiment, when the folding unlocking member **252** is in the first position, the operation portion **2512** of the operation member **251** is positioned above both the horizontal plane where the first pivot shaft **2201** is located and the horizontal plane where the second pivot shaft **2501** is located, and a first angle $\theta 1$ between the plane defined collectively by the first pivot shaft **2201** and the second pivot shaft **2501** (i.e., a first plane **S1**) and the vertical direction is an acute angle. In some embodiments not shown, the first angle $\theta 1$ may be a right angle or an obtuse angle, which is not limited in the present disclosure. Referring again to FIG. **11**, in some embodiments, when the folding unlocking member **252** is in the first position, the plane defined collectively by the operation portion **2512** and the second pivot shaft **2501** (i.e., a second plane **S2**) is coplanar with the plane defined collectively by the first pivot shaft **2201** and the second pivot shaft **2501**. It is to be understood that the plane defined collectively by the first pivot shaft **2201** and the second pivot shaft **2501** refers to a plane where both the pivot axis **R2** of the first pivot shaft **201** and the pivot axis **R3** of the second pivot shaft **501** are located, and the plane defined collectively by the operation portion **2512** and the second pivot shaft **2501** refers to a plane where both the substantially middle of the operation portion **2512** and the pivot axis **R3** of the second pivot shaft **2501** are located.

[0161] As the folding unlocking member **252** rotates from the first position to the second position, the folding unlocking member **252** rotates relative to the second pivoting base **223** of the folding joint **2200** to unlock the folding joint **2200**. When the folding unlocking member **252** rotates to the second position, both the folding joint **2200** and the connecting joint **2100** are unlocked to allow for the folding of the child carrier **2000**. Additionally, when the folding unlocking member **252** rotates to the second position, the second angle $\theta 2$ formed between the first plane **S1** and the vertical direction is an acute angle, and the second angle $\theta 2$ is smaller than the first angle $\theta 1$. At this point, the second plane **S2** is parallel to the direction of the pulling force **F** applied to the operation member **251**, and the direction of the pulling force **F** is upward directed in the vertical direction.

[0162] As the operation member **251** is further pulled, the operation member **251** drives the folding unlocking member **252** to rotate from the second position to the third position around the first pivot shaft **2201** via the second pivot shaft **2501** under the action of the pulling force **F**. When the folding unlocking member **252** is in the third position, the first plane **S1** is parallel to the direction of the pulling force **F**, the pulling force **F** is upward directed in the vertical direction, and the folding joint **2200** is in the unlocked state. During the rotation of the folding unlocking member **252** from the second position to the third position, the second plane **S2** remains parallel to the direction of the pulling force **F**, the pulling force **F** remains upward directed in the vertical direction, and both the folding joint **2200** and the connecting joint **2100** are in the unlocked state, so that the child carrier **2000** can be automatically folded under the action of gravity.

[0163] In order to reduce the volume of the child carrier **2000** or the child carrying device **2400** that is folded, the operation member **251** is rotated in a direction toward the second side wall **25232** of the limiting recess **2523** (i.e., a direction of the rotation of the folding unlocking member **252** from the first position to the third position) to abut against the inner wall of the second side wall **25232**, so that the operation member **251** is deflected away from the vertical direction. As such, the volume of the child carrying device **2400** in the vertical direction is reduced. In some embodiments, the second sidewall **25232** is inclined to the first plane **S1**, and the distance between the second

sidewall **25232** and the first plane **S1** decreases from a position close to the opening end of the limiting recess **2523** to a position away from the opening end of the limiting recess **2523**.

[0164] The folding operation of the child carrier **2000** is briefly described below with reference to FIG. **11** to FIG. **16**.

[0165] To fold the child carrier **2000**, the backrest tube **242** is firstly pushed forward and downward such that the backrest tube **242** drives the folding unlocking member **252** to rotate along with it due to the concave-convex fit between the first cooperation portion and the second cooperation portion. During this process, the operation member **251** rotates with the folding unlocking member **252** since the operation member **251** is blocked by the first side wall **25231**. Referring also to FIG. **11**, FIG. **15**, FIG. **19**, and FIG. **20**, when the backrest tube **242** is folded in place, the folding unlocking member **252** is in the first position, the push post **2271** of the push member **227** is aligned with the abutment post **2291** of the movable piece **229**, and the folding joint **2200** is in the locked state. Then a pulling force **F** is applied to the operation portion **2512** of the operation member **251** to cause the operation member **251** to rotate relative to the folding unlocking member **252**, while the operation member **251** drives the folding unlocking member **252** to rotate relative to the second pivoting base **223** and the mounting base **221** toward the second position via the second pivot shaft **2501**. During the rotation of the folding unlocking member **252** toward the second position, the push member **227** moves in the axial direction such that the push post **2271** thereof pushes the abutment post **2291** of the movable piece **229** to move the movable piece **229** in the axial direction, and the movable piece **229** in the axial movement drives the folding locking member **228** to move toward the unlocking position. Referring to FIG. **12** and FIG. **16**, when the folding unlocking member **252** is in the second position, the at least one first convex portion **2521** of the first cooperation portion abuts against the at least one second convex portion **2272** of the second cooperation portion, and the push member **227** drives the folding locking member **228** to move to the unlocking position via the movable piece **229**, allowing the relative rotation of the first pivoting base **222** and the mounting base **221**, so that the child carrying device **2400** can be further folded. Additionally, during the process of driving the folding locking member **228** to move toward the unlocking position, the protruding finger **2281** on the folding locking member **228** pushes against the first release member **214**. Referring to FIG. **15** and FIG. **16**, it can be seen in conjunction with the above that the first release member **214** pushed by the folding locking member **228** rotates around the first shaft **21211** to directly or indirectly drive the detent member **213** to move to the release position, thereby releasing the lock between the first connecting base **211** and the second connecting base **212** and allowing the folding of the supporting device **2300**.

[0166] As the operation portion **2512** of the operation member **251** is further pulled upward in the vertical direction, the operation member **251** drives the folding unlocking member **252** to rotate upward around the first pivot shaft **2201** via the second pivot shaft **2501**, and the supporting device **2300** of the child carrier **2000** is folded under gravity, during which the detent member **213** remains in the release position. At the same time, the first pivoting base **222** and the second pivoting base **223** of the child carrying device **2400** rotates downwardly around the first pivoting shaft **2201** under gravity, such that the child carrier **2000** is further folded.

[0167] Referring to FIG. **13**, FIG. **17**, and FIG. **18**, when the child carrier **2000** is folded in place, the folding unlocking member **252** rotates to the third position, with the pivot axis **R3** defined by the second pivot shaft **2501** being substantially directly above the pivot axis **R2** defined by the first pivot shaft **2201** (i.e. the first plane **S1** is substantially parallel to the vertical direction). Thereafter, the vertically upward pulling force **F** can be released, and the operation member **251** can rotate toward the second side wall **25232** of the limiting recess **2523** to abut against the second side wall **25232** of the limiting recess **2523**, which can further reduce the volume of the child carrier **2000** that is folded (specifically, the longitudinal volume of the child carrier **2000** that is folded). Upon completion of the folding of the child carrier **2000**, the folding locking member **228** is blocked by the end of the first inner gear ring **2224** of the mounting base **221** and is limited to the unlocking

position, and the detent member **213** is blocked by the end of the second inner gear ring **2123** and is limited to the release position. In other words, after the child carrier **2000** is folded in place, the supporting device **2300** is able to switch to the unfolded state, and the child carrying device **2400** is also able to switch to the unfolded state. In addition, after the child carrier **2000** is folded, when the push member **227** moves toward the folding unlocking member **252** along the first pivot shaft **2201** and is reset under the force of the resilient reset member **2232**, the second cooperation portion of the push member **227** is in convex-concave fit with the first cooperation portion of the folding unlocking member **252**.

[0168] The unfolding operation of the child carrier **2000** is briefly described below with reference to FIG. **11** to FIG. **16**.

[0169] As can be seen from the above, after the child carrier **2000** is folded in place, both the folding joint **2200** of the child carrying device **2400** and the connecting joint **2100** of the supporting device **2300** are in the unlocked state, allowing the supporting device **2300** and the child carrying device **2400** to switch to the unfolded state.

[0170] Firstly, an external force is applied to the handlebar **231** such that the handlebar **231** drives the first connecting base **211** to rotate relative to the second connecting base **212** to switch the supporting device **2300** from the folded state to the unfolded state (see FIG. **12**). At this point, as shown in FIG. **16** and FIG. **18**, the child carrying device **2400** remains in the folded state, and the at least one protruding finger **2281** of the folding locking member **228** abuts against the first release member **214** to indirectly restrict the detent member **213** in the release position. Then, the backrest tube **242** is rotated upward to drive the second pivoting base **223** to rotate with it relative to the first pivoting base **222** and the mounting base **221**. During this process, the push member **227** rotates with the second pivoting base **223** and misaligned with the abutment post **2291** of the movable piece **229**. With the second cooperation portion of the push member **227** in concave-convex fit with the first cooperation portion of the folding unlocking member **252**, along with force of the resilient reset member **2232** applied to the push member **227**, the push member **227** drives the folding unlocking member **252** to rotate with the second pivoting base **223**.

[0171] The backrest tube **242** is further rotated upward to drive the first pivoting base **222** and the folding unlocking member **252** to rotate together relative to the mounting base **221** via the second pivoting base **223**. Based on the above discussion, it can be known that when the second pivoting base **223** rotates to the extent that the first limiting tooth grooves **22241** are aligned with the respective second limiting tooth grooves **22141**, the limiting tooth **22831** is no longer blocked by the first inner gear ring **2224**, and the folding locking member **228** is reset to the locking position under the action of the force of the resilient reset member **226**, which causes detent member **213** to move axially and reset to the detent position under the action of the force of the resilient reset member **215** to lock the supporting device **2300** in the unfolded state. During the axial movement of the detent member **213**, the detent member **213** pushes the drive member **216** to move and reset, the drive member **216** pushes the first release member **214** to rotate and reset, and the first release member **214** pushes the second release member **217** to rotate and reset, thereby locking the supporting device **2300** in the unfolded state.

[0172] In the child carrier and the child carrying device provided according to the second embodiment of the present disclosure, the operation member is a rigid member and is pivotably connected to the folding unlocking member via the second pivot shaft that is offset from the first pivot shaft. As compared to the coaxial arrangement of the second pivot shaft and the first pivot shaft, the offset of the second pivot shaft from the first pivot shaft in the present disclosure can increase the torque corresponding to the force applied by the user to the operation member, thus making the unlocking operation more labor-saving. In addition, by pivotably connecting the rigid operation member to the folding unlocking member, the failure of the operation member to transmit sufficient driving force to the folding joint due to the deformation and skewing of the operation member when it is operated can be avoided. Due to the rigid structure of the operation member, the

direction of the force applied to the operation member can be well controlled, so that the operation member can more effectively transmit the driving force to the folding joint, and the situation where the folding joints at two sides are subjected to unevenly distributed loads which causes asynchronized unlocking of the folding joints can be avoided, thereby achieving the rapid folding of the child carrier **2000** in a more efficient manner with less effort.

Third Embodiment

[0173] The child carrier, such as a stroller, is typically designed with a folding function that minimizes the space it occupies, making it easier to store and transport. However, it is difficult for the conventional child carrier to control the folding speed during the folding process, which can lead to violent impacts between the frame components, potentially causing damage to the stroller.

[0174] In view of this, a frame and a child carrier including the same are provided according to a third embodiment of the present disclosure, as shown in FIG. 23 to FIG. 28.

[0175] Specifically, as shown in FIG. 23, a child carrier (e.g., a stroller) **3000** is provided according to the third embodiment of the present disclosure, which includes a frame **310**. The frame **310** includes a first frame body **3100**, a second frame body **3200**, a third frame body **3700**, a linkage frame body **316**, and a locking joint **314**. The first frame body **3100** is pivotally connected to the second frame body **3200**. The locking joint **314** is connected between the first frame body **3100** and the third frame body **3700**, and the first frame body **3100** is pivotally connected to the third frame body **3700** by the locking joint **314**. The linkage frame body **316** is pivotally connected to each of the second frame body **3200** and the third frame body **3700**. The locking joint **314** has a locked state and an unlocked state. When the locking joint **314** is in the locked state, the first frame body **3100** and the third frame body **3700** are locked to each other such that they are unable to rotate relative to each other, and the frame **310** is in the unfolded state as shown in FIG. 23. When the locking joint **314** is in the unlocked state, the first frame body **3100** and the third frame body **3700** are able to rotate relative to each other, and the frame **310** can be switched from the unfolded state to the folded state. In this embodiment, the pivoting position of the first frame body **3100** and the second frame body **3200** is arranged off-axis from the locking joint **314**, i.e., the second frame body **3200** and the third frame body **3700** are pivotally connected to the first frame body **3100** at different positions of the first frame body **3100**, respectively.

[0176] For ease of understanding, a coordinate system is established in the drawings. As shown in FIG. 23, the frame **310** is connected to the front wheel assembly **318a** and the rear wheel assembly **318b**. When the frame **310** is placed on a horizontal plane, it is defined that the front wheel assembly **318a** is located at the front side of the rear wheel assembly **318b**, i.e., the front wheel assembly **318a** is located in a negative direction of the X-axis with respect to the rear wheel assembly **318b**. The positive direction of the Z-axis is parallel to the vertically upward direction. Two locking joints **314** are located in a line parallel to the direction of the Y-axis, and the locking joint **314** at the right side is located in a positive direction of the Y-axis with respect to the locking joint **314** at the left side.

[0177] Referring to FIG. 23, FIG. 26, and FIG. 27, during the switching process of the frame **310** from the unfolded state to the folded state, the first frame body **3100** and the second frame body **3200** get closer to each other, and the linkage frame body **316** drives the third frame body **3700** and the second frame body **3200** to get closer to each other. More specifically, during the switching process of the frame **310** from the unfolded state to the folded state, taking the second frame body **3200** as a reference, the first frame body **3100** rotates backward and downward toward the second frame body **3200**, and the third frame body **3700** rotates backward toward the second frame body **3200**. As shown in FIG. 27, when the frame **310** is in the folded state, the first frame body **3100** is located behind the second frame body **3200**, and the third frame body **3700** is located in front of the second frame body **3200**. In other embodiments, the rotation direction between the first frame body **3100**, the second frame body **3200**, and the third frame body **3700** may vary, for example, the first frame body **3100** rotates forward and downward toward the third frame body **3700**.

[0178] It should be noted that the relative rotation between the first frame body **3100** and the second frame body **3200** can exhibit different rotation directions depending on the chosen reference method. For example, in this embodiment, during the switching process of the frame **310** from the unfolded state to the folded state, the relative rotation between the first frame body **3100** and the second frame body **3200** may be viewed as either the rotation of the second frame body **3200** relative to the first frame body **3100** in the first rotation direction **A1** or the rotation of the first frame body **3100** relative to the second frame body **3200** in the second rotation direction **A2** opposite to the first rotation direction **A1**. Taking the perspective of FIG. **26** as an example, the first rotation direction **A1** corresponds to the counterclockwise direction of FIG. **26**, while the second rotation direction **A2** corresponds to the clockwise direction of FIG. **26**. For ease of understanding, in the following, the relative rotation between the first frame body **3100** and the second frame body **3200** will be described mainly with the rotation of the second frame body **3200** relative to the first frame body **3100** as an example.

[0179] Referring to FIG. **24** to FIG. **26**, the frame **310** further includes a rotation member **3300** and a buffer member **3400**. The rotation member **3300** is pivotally connected to the first frame body **3100**, i.e., the rotation member **3300** is rotatable relative to the first frame body **3100**. The buffer member **3400** is connected to the rotation member **3300**. The buffer member **3400** is configured to apply a force to the rotation member **3300** intended to rotate the rotation member **3300** relative to the first frame body **3100** in the second rotation direction **A2** opposite to the first rotation direction **A1**. As the second frame body **3200** rotates relative to the first frame body **3100** in the first rotation direction **A1**, the frame **310** switches from the unfolded state to the folded state. During the rotation of the second frame body **3200** relative to the first frame body **3100** in the first rotation direction **A1**, the second frame body **3200** drives the rotation member **3300** to rotate relative to the first frame body **3100** in the first rotation direction **A1**. In this embodiment, the rotation member **3300** has a pivot end **3310** pivotally connected to the first frame body **3100** and a cooperation end **3320** cooperating with the second frame body **3200**. The cooperation end **3320** and the pivot end **3310** are located at opposite ends of the rotation member **3300**. In this embodiment, the cooperation end **3320** cooperates with and abuts against the second frame body **3200**. The second frame body **3200** drives the rotation member **3300** to rotate relative to the first frame body **3100** in the first rotation direction **A1** by pushing against the cooperation end **3320**.

[0180] It should be understood that when the second frame body **3200** rotates relative to the first frame body **3100** in the first rotation direction **A1**, the second frame body **3200** drives the rotation member **3300** to rotate relative to the first frame body **3100** in the first rotation direction **A1**. However, since the buffer member **3400** apply a force to the rotation member **3300** intended to rotate the rotation member **3300** relative to the first frame body **3100** in the second rotation direction **A2**, the buffer member **3400** has a damping effect on the rotation of the rotation member **3300** relative to the first frame body **3100** in the first rotation direction **A1**, which increase the resistance to the rotation of the rotation member **3300** relative to the first frame body **3100** in the first rotation direction **A1**, thereby reducing the speed of the rotation of the rotation member **3300** relative to the first frame body **3100** in the first rotation direction **A1**. As such, a violent collision between the first frame body **3100** and the second frame body **3200** can be avoided during the folding process of the frame **310**, or a collision between other components driven by the first frame body **3100** and the ground or other structures can be avoided.

[0181] Referring to FIG. **25** to FIG. **28**, a first angle α is formed between the first frame body **3100** and the second frame body **3200**. During the folding process of the frame **310**, the first angle α gradually decreases, and the rotation member **3300** is positioned in an angular area of the first angle α (i.e., an area corresponding to the first angle α). More specifically, the first frame body **3100** has a first edge **3121** and a second edge **3122** opposite to each other. The first edge **3121** is located at the side of the first frame body **3100** facing the rotation member **3300**. The second frame body **3200** has a third edge **3214** and a fourth edge **3215** opposite to each other. The third edge

3214 is located at the side of the second frame body **3200** facing the rotation member **3300**. The first angle α is formed between the first edge **3121** and the third edge **3214**. A value of the first angle α when the frame **310** is in the unfolded state is greater than a value of the first angle α when the frame **310** is in the folded state. In other words, when the frame **310** is in the folded state, the first edge **3121** is closer to the third edge **3214** than the second edge **3122**, while the third edge **3214** is closer to the first edge **3121** than the fourth edge **3215**. The rotation member **300** is located in the area between the first edge **3121** and the third edge **3214**. During the rotation of the second frame body **3200** relative to the first frame body **3100** in the first rotation direction **A1**, the rotation member **3300** abuts against the third edge **3214** of the second frame body **3200**.

[0182] Referring to FIG. 23 and FIG. 24, the first frame body **3100** includes a first main frame body **3120** and a first connecting base **3110** that are connected to each other and arranged at an angle. The first edge **3121** and the second edge **3122** of the first frame body **3100** are two edges of the first main frame body **3120**. The second frame body **3200** includes a second main frame body **3213** and a second connecting base **3210** that are connected to each other. The first connecting base **3110** is pivotably connected to the second connecting base **3210**. As shown in FIG. 25, when the frame **310** is in the unfolded state, at the position of the cooperation end **3320** of the rotation member **3300**, an edge of the first connecting base **3110** facing the rotation member **3300** is closer to the cooperation end **3320** of the rotation member **3300** than an edge of the second connecting base **3210** facing the rotation member **3300**. Therefore, when the frame **310** is in the unfolded state, the rotation member **3300** abuts against the edge of the first connecting base **3110** facing the rotation member **3300**. In alternative embodiments, at the position of the cooperation end **3320** of the rotation member **3300**, the edge of the second connecting base **3210** facing the rotation member **3300** is closer to the cooperation end **3320** of the rotation member **3300** than the edge of the first connecting base **3110** facing the rotation member **3300**, and in this case, when the frame **310** is in the unfolded state, the rotation member **3300** abuts against the edge of the second connecting base **3210** facing the rotation member **3300**.

[0183] Referring to FIG. 24 to FIG. 28, during the switching process of the frame **10** from the unfolded state to the folded state, the edge of the second connecting base **3210** facing the rotation member **3300** pushes against the rotation member **3300** to rotate relative to the first frame body **3100** in the first direction of rotation **A1**. More specifically, in this embodiment, the second main frame body **3213** and the second connecting base **3210** extend in substantially the same direction, and the second connecting base **3210** protrudes from an end of the second main frame body **3213**. The third edge **3214** includes a first section **3214a** and a second section **3214b**. The first section **3214a** of the third edge **3214** of the second frame body **3200** is located on the second connecting base **3210**, while the second section **3214b** of the third edge **3214** is located on the second main frame body **3213**. A first section of the fourth edge **3215** of the second frame body **3200** is located on the second main frame body **3213**, while a second section of the fourth edge **3215** is located on the second connecting base **3210**. During the rotation of the second frame body **3200** relative to the first frame body **3100** in the first rotational direction **A1**, the cooperation end **3320** of the rotation member **3300** gradually slides from the second connecting base **3210** to the second main frame body **3213**, and the rotation member **3300** firstly abuts against the edge of the second connecting base **3210** facing the rotation member **3300** (i.e., the first section **3214a** of the third edge **3214**) and then abuts against the edge of the second main frame body **3213** facing the rotation member **3300** (i.e., the second section **3214b** of the third edge **3214**). In other embodiments, if the length of the second connecting base **3210** protruding from the end of the second main frame body **3213** is sufficiently long, the rotation member **3300** may abut against and slide along only the edge of the second connecting base **3210** facing the rotation member **3300** (i.e., the first section **3214a** of the third edge **3214**); alternatively, if the extension length of the rotation member **3300** is relatively long, the rotation member **3300** may abut against and slide along only

the edge of the second main frame body **3213** facing the rotation member **3300** (i.e., the second section **3214b** of the third edge **3214**).

[0184] Further, referring to FIG. **24**, the first connecting base **3110** has a first base body **3111** and a second base body **3112** spaced apart from each other. In this embodiment, the first base body **3111** and the second base body **3112** are of the same shape, both being approximately of a flattened block structure. The first base body **3111** and the second base body **3112** are spaced apart from each other in the Y-axis direction. The first connecting base **3110** also has a cavity **3114** and an opening **3115** in communication with each other. The cavity **3114** is located between the first base body **3111** and the second base body **3112**. At least a part of the second connecting base **3210** is embedded in the cavity **3114**. The opening **3115** is disposed opposite to the rotation member **3300** and the edge of the second connecting base **3210** facing the rotation member **3300** (i.e., the first section **3214a** of the third edge **3214**) is exposed through the opening **3115**. The first base body **3111** and the second base body **3112** each have a side edge facing the rotation member **3300**. More specifically, side edges of the first base body **3111** and the second base body **3112** corresponding to the positive direction of the X-axis face the rotation member **3300**. In this embodiment, as shown in FIG. **23** and FIG. **25**, when the frame **310** is in the unfolded state, the cooperation end **3320** of the rotation member **3300** abuts against at least one of the edges of the first base body **3111** and the second base body **3112** facing the rotation member **3300**. As shown in FIG. **26** and FIG. **27**, during the switching process of the frame **310** from the unfolded state to the folded state, the edge of the second connecting base **3210** facing the rotation member **3300** (i.e., the first section **3214a** of the third edge **3214**) extends out of the opening **3115** and pushes the rotation member **3300** to rotate relative to the first frame body **3100** in the first rotational direction **A1**. Further, as shown in FIG. **24** and FIG. **26**, the first connecting base **3110** further includes a connecting portion **3113** connecting the first base body **3111** to the second base body **3112**. The cavity **3114** is defined by the connecting portion **3113**, the first base body **3111**, and the second base body **3112**.

[0185] It should be noted that in this embodiment, the connecting portion **3113** is connected to the first main frame body **3120** (e.g., by fasteners such as screws). In other embodiments, the first connecting base **3110** may not be provided with the connecting portion **3113**, the first base body **3111** and the second base body **3112** may each be directly connected to the first main frame body **3120** by welding, gluing, or other methods, and the cavity **3114** may be defined by the first base body **3111**, the second base body **3112**, and the first edge **3121** of the first main frame body **3120**. In yet other embodiments, the first connecting base **3110** of the first frame body **3100** may be omitted, and the second frame body **3200** may be directly pivotably connected to the first main frame body **3120**.

[0186] It should be noted that in this embodiment, the second connecting base **3210** and the second main frame body **3213** may be separate components and mounted to each other to achieved the connection therebetween, for example, a part of the second connecting base **3210** is inserted into the second main frame body **3213**. In other embodiments, the second connecting base **3210** and the second main frame body **3213** may be integrally formed, which is not limited herein.

[0187] Referring to FIG. **24**, FIG. **25**, and FIG. **26**, a gap exists at the pivot position between the first frame body **3100** and the second frame body **3200** and is obscured by the rotation member **3300**. Specifically, there is a gap between the second connecting base **3210** and the first base body **3111**, and there is also a gap between the second connecting base **3210** and the second base body **3112**. The rotation member **3300** covers the gaps to prevent the user's fingers and other parts of the user from extending into the gaps, thereby preventing the user from being injured during the folding process of the frame **310** and reducing the risk of pinching the user's hand. Specifically, a width (i.e., a dimension along the Y-axis) of the rotation member **3300** is greater than a distance between the first base body **3111** and the second base body **3112**, and the rotation member **3300** covers at least a part of the opening **3115**.

[0188] Referring to FIG. **24** to FIG. **26**, the frame **310** further includes a first pivot shaft **3500**. The

first connecting base **3110** is pivotally connected to the second connecting base **3210** via the first pivot shaft **3500**. More specifically, the first connecting base **3110** is provided with a first pivot hole **3116** (the first base body **3111** and the second base body **3112** are each provided with a first pivot hole **3116**). The first pivot hole **3116** extends through the first connecting base **3110** along the thickness direction of the first connecting base **3110**, i.e., a direction parallel to the Y-axis. The second connecting base **3210** is provided with a second pivot hole **3212**. The second pivot hole **3212** is located near a middle portion of the second connecting base **3210** and extends through the second connecting base **3210** along the thickness direction of the second connecting base **3210**, i.e., a direction parallel to the Y-axis. The first pivot shaft **3500** is disposed through the first pivot hole **3116** and the second pivot hole **3212** such that the second connecting base **3210** is rotatably connected to the first connecting base **3110**.

[0189] Further, the first connecting base **3110** has a first portion **3110a** between the connection position of the first connecting base **3110** to the first main frame body **3120** and the first pivot shaft **3500**. The first connecting base **3110** also has a second portion **3110b** between the first pivot shaft **3500** and the end of the first connecting base **3110**. To facilitate understanding of the first portion **3110a** and the second portion **3110b** of the first connecting base **3110**, the first portion **3110a** and the second portion **3110b** are divided by a dotted line in FIG. 23 and FIG. 24. However, in fact, for the integrally formed first connecting base **3110**, there is no obvious boundary between the first portion **3110a** and the second portion **3110b** in appearance. It should be understood that the position and the line type of the dotted line may vary, and the shapes and sizes of the first portion **3110a** and the second portion **3110b** are not limited by the dotted lines in FIG. 23 and FIG. 24. When the frame **310** is in the unfolded state, the rotation member **3300** shield the area of the opening **3115** that corresponds to the first portion **3110a**. The first portion **3110a** of the first connecting base **3110** is closer to the connection position between the first connecting base **3110** and the first main frame body **3120** than the second portion **3110b** of the first connecting base **3110**. Additional, during the folding process, the gap between the second connecting base **3210** and the opening **3115** of the first connecting base **3110** becomes larger and larger, and the risk of pinching user's finger can be effectively reduced by shielding the area of the opening **3115** that corresponds to the first portion **3110a** of the first connecting base **3110** by the rotation member **3300**. It should be understood that in other embodiments, the rotation member **3300** may further extend to shield the entire opening **3115**.

[0190] Further, referring again to FIG. 24, the frame **310** further includes a second pivot shaft **3600**. The rotation member **3300** is pivotally connected to the first frame body **3100** via the second pivot shaft **3600**. In this embodiment, the rotation member **3300** is pivotally connected to first connecting base **3110** via the second pivot shaft **3600**, the first connecting base **3110** is provided with a third pivot hole **3119** on a side thereof corresponding to the positive direction of the X-axis, the rotation member **3300** is provided with a fourth pivot hole **3313**, and the second pivot shaft **3600** is disposed through the third pivot hole **3119** and the fourth pivot hole **3313**. More specifically, the first connecting base **3110** is provided with a first protruding portion **3117** and a second protruding portion **3118**, and the rotation member **3300** is provided with a third protruding portion **3311** and a fourth protruding portion **3312**. Referring to FIG. 24, specifically, the first protruding portion **3117** and the second protruding portion **3118** are both provided on one end of the connecting portion **3113** in the position direction of the X-axis, and the first protruding portion **3117** and the second protruding portion **3118** are spaced apart from each other in the Y-axis direction. The third protruding portion **3311** and the fourth protruding portion **3312** are also spaced apart from each other in the Y-axis direction, and are both provided on the pivot end **3310** of the rotation member **3300**, i.e., the end of the rotation member **3300** in the positive direction of the Z-axis. The first protruding portion **3117** and the second protruding portion **3118** are each provided with a third pivot hole **3119**, and the third protruding portion **3311** and the fourth protruding portion **3312** are each provided with a fourth pivot hole **3313**. The first protruding portion **3117** and the

second protruding portion **3118** are located between the third protruding portion **3311** and the fourth protruding portion **3312**. The second pivot shaft **3600** is disposed through the third protruding portion **3311**, the first protruding portion **3117**, the second protruding portion **3118**, and the fourth protruding portion **3312** to pivotably connect the rotation member **3300** to the first connecting base **3110**. In other embodiments, the rotation member **3300** may be pivotally connected to first main frame body **3120**, i.e., the pivot end **3310** of the rotation member **3300** may be pivotally connected to first main frame body **3120**, which is not limited herein.

[0191] In some embodiments, the buffer member **3400** is a resilient member. The resilient member may be a torsion spring, a spring, a rubber member, and the like. When the second connecting base **3210** rotates relative to the first connecting base **3110** in the first direction of rotation **A1**, the rotation member **3300** abuts against the second frame body **3200** and rotates with the second frame body **3200** in the first direction of rotation **A1**, causing the resilient member to be deformed and generate a reset force which is applied to the rotation member **3300** such that the rotation member **3300** has a tendency to rotate in the second direction of rotation **A2**. Thus, when the frame **310** switches from the folded state to the unfolded state, the rotation member **3300** is automatically rotated along the second rotational direction **A2** by the elastic member, without the need for the user to manually reset the rotation member **3300**, improving the convenience of use.

[0192] Further, the buffer member **3400** in this embodiment is a torsion spring **3410**. In this embodiment, as shown in FIG. 24 and FIG. 26, the torsion spring **3410** includes two torsion arms **3411** and a body **3412**. One of the torsion arms **3411** is connected to the first frame body **3100** (e.g., fixedly connected to the connection portion **3113** of the first connecting base **3110**). The rotation member **3300** has an accommodation groove **3330** (see FIG. 25), and the other one of the torsion arms **3411** of the torsion spring **3410** is disposed in the accommodation groove **3330** (i.e., the other one of the torsion arms **3411** of the torsion spring **3410** is inserted into the accommodation groove **3330**). In this way, the accommodation groove **3330** can effectively prevent the torsion arm **3411** from being disconnected from the rotation member **3300**, improve the stability of the connection between the torsion spring **3410** and the rotation member **3300**, and can more stably apply a reset force to the rotation member **3300** to prevent the reset from failing. In addition, the first protruding portion **3117** and the second protruding portion **3118** are spaced apart from each other, which can not only reserve installation space for the body **3412** of the torsion spring **3410**, but also limit the body **3412**, further improving the installation stability of the torsion spring **3410**, and preventing the torsion spring **3410** from coming loose.

[0193] Referring again to FIG. 23, FIG. 25 and FIG. 28, the frame **310** is placed on a horizontal plane (not shown), and the frame **310** includes a handlebar frame **311**, a front leg frame **312**, a rear leg frame **313**, a push handle **315**, a linkage frame **316**, and a footrest frame **317**. There are two handlebar frames **311** each disposed inclined relative to the horizontal plane. There are two front leg frames **312** each disposed inclined relative to the horizontal plane. The front leg frames **312** extend in substantially the same direction as the handlebar frames **311**, respectively. The push handle **315** is of a substantially U-shaped rod structure, with two ends connected to the upper ends of the two handlebar frames **311**, respectively. The lower ends of the two handlebar frames **311** are connected to the upper ends of the two front leg frames **312** via two locking joints **314**, respectively. The lower ends of the two front leg frames **312** are connected to the ends of the footrest frame **317**, respectively. Two front wheel assemblies **318a** are connected to the bottom of the footrest frame **317**. There are two rear leg frames **313** disposed inclined relative to the horizontal plane. The upper ends of the two rear leg frames **313** are pivotally connected to the two handlebar frames **311**, respectively. The lower ends of the two rear leg frames **313** are connected to two rear wheel assemblies **318b**, respectively. When the frame **310** is in the unfolded state, the handlebar frames **311**, the front leg frames **312**, the footrest frame **317**, and the push handle **315** are substantially located in the same plane and collectively form a substantially rectangular structure. It should be understood that in other embodiments, the shapes and numbers of the above components

of the frame **310** may vary depending on actual needs, and are not specially limited herein.

[0194] In this embodiment, the first frame body **3100** is the handlebar frame **311** of the frame **310**, the second frame body **3200** is the rear leg frame **313** of the frame **310**, and the third frame body **3700** is the front leg frame **312** of the frame **310**. The rotation member **3300** is located at the side of the rear leg frame **313** facing the push handle **315**, i.e., the rotation member **3300** is located at the side of the rear leg frame **313** corresponding to the positive direction of the X-axis. The provision of the rotation member **3300** and the buffer member **3400** can slow down the folding between the handlebar frame **311** and the rear leg frame **313**, thereby preventing the violent collision between the handlebar frame **311** and the rear leg frame **313**. In addition, as shown in FIG. **28**, when the frame **310** is in the folded state, the push handle **315** abuts against the support surface (e.g., the ground) to assist the folded frame **310** to remain standing. In the present disclosure, by slowing down the rotation the second frame body **3200** relative to the first frame body **3100** in the first rotation direction **A1**, the rotation of the push handle **315** driven by the first frame body **3100** can also be slowed down, thereby preventing the violent collision between the push handle **315** and the support surface which may cause damage to the frame **310**.

[0195] In other embodiments, the first frame body **3100** and the second frame body **3200** may be other components of the frame **310** that need to slow down their rotation and to prevent the pinching of user's hand. For example, the frame **310** further includes an armrest frame (or dinner plate holder) that may be folded toward the front leg frame **312**, and the first frame body **3100** and the second frame body **3200** may be the front leg frame **312** and the armrest frame (or dinner plate holder) of the frame **310**, respectively.

[0196] In the above frame, the buffer member has a damping effect on the rotation of the rotation member relative to the first frame body in the first rotation direction, which increase the resistance to the rotation of the rotation member relative to the first frame body in the first rotation direction, thereby slowing down the rotation of the second frame body relative to the first frame body in the first rotation direction. As a result, the violent collision between the first frame body and the second frame body in the folding process of the frame can be prevented, or the collision between other components driven by the first frame body and the ground or other structures can be prevented, thereby reducing the risk of damaging the frame.

[0197] The technical features of the above-mentioned embodiments can be combined arbitrarily. In order to make the description concise, not all possible combinations of the technical features are described in the embodiments. However, as long as there is no contradiction in the combination of these technical features, the combinations should be considered as in the scope of the present disclosure.

[0198] The above-described embodiments are only several implementations of the present disclosure, and the descriptions are relatively specific and detailed, but they should not be construed as limiting the scope of the present disclosure. It should be understood by those of ordinary skill in the art that various modifications and improvements can be made without departing from the concept of the present disclosure, and all fall within the protection scope of the present disclosure. Therefore, the patent protection of the present disclosure shall be defined by the appended claims.

Claims

1. A connecting structure, comprising: a first joint component and a second joint component pivotable relative to each other around an axial direction; and a connecting member extending through the second joint component in the axial direction and inserted into and screwedly connected to the first joint component to restrict a separation of the first joint component from the second joint component in a direction parallel to the axial direction; wherein the connecting member is spaced apart from the second joint component.

2. The connecting structure of claim 1, wherein the first joint component is positioned between the connecting member and the second joint component in a radial direction perpendicular to the axial direction to space apart the connecting member from the second joint component in the radial direction.
3. The connecting structure of claim 2, wherein the first joint component comprises a shaft portion extending through the second joint component in the direction parallel to the axial direction, and the shaft portion is positioned between the connecting member and the second joint component in the radial direction.
4. The connecting structure of claim 3, wherein the connecting member is provided with a head portion and a rod portion extending from the head portion in the axial direction, the rod portion is inserted into the shaft portion along the axial direction and threadedly connected to the shaft portion, the head portion is positioned at an outer side of the shaft portion in the axial direction, and the second joint component is positioned between the head portion and the first joint component in the direction parallel to the axial direction.
5. The connecting structure of claim 3, wherein the connecting member further comprises a spacer member, the connecting member is provided with a head portion and a rod portion extending from the head portion in the axial direction, the rod portion is inserted into the shaft portion along the axial direction and threadedly connected to the shaft portion, the head portion is positioned at an outer side of the shaft portion in the axial direction, the spacer member is positioned between the head portion and the second joint component in the direction parallel to the axial direction to space apart the connecting member from the second joint component in the direction parallel to the axial direction, and the spacer member is unable to rotate relative to the connecting member around the axial direction.
6. The connecting structure of claim 5, wherein the spacer member is sleeved outside of the shaft portion along the direction parallel to the axial direction, and the spacer member is unable to rotate relative to the shaft portion around the axial direction.
7. The connecting structure of claim 6, wherein an inner peripheral surface of the spacer member is positioned adjacent to an outer surface of the shaft portion in the radial direction, and a shape of a cross section of the inner peripheral surface of the spacer member taken along a direction perpendicular to the axial direction and a shape of a cross section of the outer surface of the shaft portion taken along the direction perpendicular to the axial direction are matched with each other and are non-circular.
8. The connecting structure of claim 5, wherein the head portion, the spacer member, the second joint component, and the first joint component abut against each other sequentially in the direction parallel to the axial direction.
9. The connecting structure of claim 5, wherein the second joint component is provided with a second recess that is recessed away from the head portion in the direction parallel to the axial direction, the spacer member is received in the second recess along the direction parallel to the axial direction, the second recess comprises a second bottom surface, and the spacer member is positioned between the head portion and the second bottom surface in the direction parallel to the axial direction.
10. The connecting structure of claim 9, wherein the second recess comprises an inner surface positioned adjacent to and spaced apart from an outer peripheral surface of the spacer member in the radial direction.
11. The connecting structure of claim 5, wherein the spacer member is positioned between the head portion and the shaft portion in the direction parallel to the axial direction and abuts against each of the head portion and the shaft portion.
12. The connecting structure of claim 5, wherein an end portion of the shaft portion proximate to the head portion is recessed away from the head portion in the direction parallel to the axial direction to form a first recess, the spacer member is received in the first recess along the direction

parallel to the axial direction, the first recess comprises a first bottom surface, and the spacer member is positioned between the head portion and the first bottom surface in the direction parallel to the axial direction.

13. The connecting structure of claim 12, wherein the first recess comprises an outer surface positioned adjacent to an inner peripheral surface of the spacer member in the radial direction, and shapes of cross sections of the inner peripheral surface of the spacer member and the outer surface taken along a direction perpendicular to the axial direction are matched with each other and are non-circular.

14. The connecting structure of claim 1, wherein the connecting structure further comprises a spacer member positioned between the connecting member and the second joint component in the direction parallel to the axial direction to space apart the connecting member from the second joint component in the direction parallel to the axial direction, and the spacer member is unable to rotate relative to the connecting member around the axial direction.

15. The connecting structure of claim 14, wherein the spacer member is sleeved outside of the first joint component along the direction parallel to the axial direction, and the spacer member is unable to rotate relative to the first joint component around the axial direction.

16. The connecting structure of claim 15, wherein an inner peripheral surface of the spacer member is positioned adjacent to an outer surface of the first joint component in the radial direction, and a shape of a cross section of the inner peripheral surface of the spacer member taken along a direction perpendicular to the axial direction and a shape of a cross section of the outer surface of the first joint component taken along the direction perpendicular to the axial direction are matched with each other and are non-circular.

17. The connecting structure of claim 14, wherein the spacer member abuts against each of the connecting member and the second joint component in the direction parallel to the axial direction, the second joint component is positioned between the connecting member and the first joint component in the direction parallel to the axial direction, and the second joint component abuts against the first joint component.

18. The connecting structure of claim 15, wherein the spacer member is positioned between the connecting member and the first joint component in the direction parallel to the axial direction, and abuts against each of the connecting member and the first joint component.

19. The connecting structure of claim 1, wherein the connecting member further comprise: a locking member disposed between the first joint component and the second joint component in the direction parallel to the axial direction and movable between a locking position and an unlocking position in the direction parallel to the axial direction, wherein in the locking position, the locking member prevents the first joint component and the second joint component from rotating relative to each other, and in the unlocking position, the locking member allows the first joint component and the second joint component to rotate relative to each other; a reset member disposed between the locking member and the first joint component and configured to bias the locking member toward the locking position; and an unlocking member movably connected to the second joint component and configured to be operable externally from the connecting structure to drive the locking member to move from the locking position to the unlocking position; wherein the locking member is positioned between the reset member and the unlocking member in the direction parallel to the axial direction.

20. The connecting structure of claim 19, wherein the connecting member is exposed outside of the first joint component and the second joint component in the axial direction, and the connecting member is positioned between the unlocking member and the first joint component and between the unlocking member and the second joint component in the direction parallel to the axial direction.
